

Economic Value of Connecting Markets: A Dynamic Spatial Equilibrium Analysis of China's Railway Expansion

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May 11, 2026

Abstract

To quantify the welfare effects of China's railway expansion from 2002 to 2016, we develop a dynamic spatial general equilibrium model featuring multi-sector production networks, forward-looking migration, and regional capital accumulation. Exploiting time-varying changes in railway distances within city pairs, we calculate the railway-related trade-cost reductions and associated changes in economic activities. Based on the quantitative model, railway expansion yields a long-run welfare gain of 9.4 percent in consumption-equivalent terms. Inland western provinces record a welfare gain of 12.3 percent, nearly double the 6.4 percent gain in the better-connected east. Input-output propagation, labor reallocation, and endogenous capital accumulation account for over four-fifths of the long-run welfare effect. The ratio of long-run to short-run gains is 5.5, implying that conventional short-horizon or static evaluations in the literature may substantially understate the returns to transport infrastructure.

Keywords: Railway infrastructure; Production networks; Internal migration; Dynamic spatial equilibrium

JEL Classification: F14, O18, R11, R13, R40

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1 Introduction

Over the fifteen years from 2002 to 2016, China’s operational railway network grew by more than 50,000 kilometers, from roughly 72,000 to over 124,000 kilometers,¹ including more than 22,000 kilometers of high-speed rail that together constitute the largest such network in the world.² Cumulative public investment in railways over this period exceeded six trillion yuan, averaging close to one percent of GDP annually at the program’s peak.³ Successive Five-Year Plans framed this effort as a tool for integrating inland provinces with coastal markets and narrowing regional income gaps, and similar arguments now motivate large-scale rail programs in India, Indonesia, Nigeria, and across the European Union. Yet despite the resources involved, the welfare returns to such investments remain difficult to quantify. How large are the aggregate gains from railway expansion? How are they distributed across regions, sectors, and workers? And through which economic channels do they propagate over time?

Answering these welfare-related questions is difficult in static and partial equilibrium frameworks because railway infrastructure operates through interacting channels that play out over very different horizons. Lower transport costs reduce delivered prices and expand the set of markets a region can reach, but they also intensify competition from previously distant suppliers. Better connectivity redirects intermediate-input flows through the production network, so that a cost reduction in one sector propagates to downstream users, whose own prices then adjust. On longer horizons, infrastructure improvements reallocate workers across regions as migration responds to the evolving distribution of real wages and amenities, and they alter investment incentives as the marginal product of capital shifts in newly connected regions. Because each margin responds with a different speed and feeds back into the others, the short-run and long-run welfare consequences of a given project can differ substantially in magnitude—and sometimes in sign. Capturing them within a

¹Data are from the National Railway Administration of China, *Statistical Bulletin on Railway Industry Development*, 2002 and 2016 issues.

²High-speed rail mileage is from the same source. The figure refers to lines operating at design speeds of 250 km/h or above, consistent with China’s official classification of high-speed rail.

³Cumulative investment is calculated by summing annual railway fixed asset investment from the *China Statistical Yearbook*, National Bureau of Statistics. GDP data are from the same source. The ratio peaks at approximately 1.2 percent of GDP in 2010–2013, coinciding with the post-crisis stimulus program.

single coherent framework requires a model in which trade, production networks, labor mobility, and capital accumulation are jointly determined in dynamic equilibrium.

To quantify the effect of railway infrastructure on aggregate welfare and the regional distribution of economic activities, as well as the importance of different propagation channels in China, we develop a dynamic spatial equilibrium model and apply it to China’s 2002–2016 railway expansion. On the theoretical side, we extend the dynamic spatial model of [Kleinman et al. \(2023\)](#) by embedding a multi-sector input-output network into an environment with forward-looking migration and region-specific capital accumulation. This extension is essential in our setting because Chinese provinces differ sharply in sectoral composition and because railway-induced cost reductions propagate through intermediate-input chains rather than remaining confined to the sectors directly affected. On the empirical side, we construct a novel prefecture-pair panel of railway shortest-path distances covering all Chinese prefecture-level administrative units from 2002 to 2016. The staggered rollout of new lines generates substantial within-pair variation in bilateral railway distance over time, which we use both to estimate the elasticity of trade costs with respect to railway connectivity and to identify the railway-induced reduction in trade costs in the structural counterfactual. Combining this network data with multi-regional input-output tables, census migration matrices, and provincial accounts, we calibrate the model through a sequential procedure that recovers each fundamental from closed-form structural inversions, and we compare the observed transition path with a counterfactual in which railway distances are held at their 2002 levels while all other fundamentals follow their actual paths.

Three sets of quantitative findings emerge. First, railway expansion generates economically meaningful welfare gains that unfold gradually: the consumption-equivalent gain rises from 1.71 percent five years out to 4.23 percent at ten years and 9.40 percent in the long run, so that only about 18 percent of the long-run effect materializes within the first five years and full convergence requires roughly three decades. Second, the gains are highly heterogeneous across space and sectors: the western and central regions gain 12.31 and 10.73 percent respectively, nearly twice the 6.42 percent gain in the already

well-connected east, while manufacturing gains 11.24 percent compared with 8.72 percent for services and 6.31 percent for agriculture, reflecting manufacturing’s higher tradability and deeper position in the input-output network. The geography of welfare gains diverges from the geography of population flows: some inland regions gain in welfare terms even as workers migrate out, because the option value of mobility itself is part of welfare. Third, decomposing the long-run gain by mechanism shows that the static price and wage channels account for only 1.71 percentage points (18.2 percent of the total), while production-network propagation contributes 2.21 points, labor migration 3.04 points, and capital accumulation 2.06 points—together more than three-quarters of the long-run gain. Shutting down any one of these dynamic margins lowers the welfare gain by roughly two to three percentage points, indicating that they are complements rather than substitutes in transmitting infrastructure shocks. Taken together, these results imply that conventional short-horizon or static evaluations understate the returns to large-scale transport infrastructure by a factor of roughly 5.5, and that the regional convergence effects long attributed to railway construction operate primarily through slow-moving factor reallocation rather than through immediate price adjustments.

This paper contributes to four strands of literature.

Quantitative spatial models. The paper builds on the now-large literature on quantitative spatial models, with foundations in [Krugman \(1991\)](#), [Eaton and Kortum \(2002\)](#), and [Anderson and van Wincoop \(2003\)](#). More recent contributions develop tractable frameworks for recovering fundamentals from spatial data and evaluating counterfactuals: [Allen and Arkolakis \(2014\)](#) and [Redding \(2016\)](#) on trade and factor mobility, [Monte et al. \(2018\)](#) on commuting, [Fajgelbaum and Gaubert \(2020\)](#) on optimal spatial policy, and [Nagy \(2023\)](#) on forward-looking dynamic adjustment. The key methodological enabler for these exercises is exact-hat algebra ([Dekle et al., 2008](#); [Caliendo and Parro, 2015](#)), and the market-access approach of [Donaldson and Hornbeck \(2016\)](#) provides a particularly useful bridge between reduced-form evidence on transport infrastructure and general equilibrium welfare analysis. Recent work incorporates dynamic labor adjustment: [Caliendo et al. \(2019\)](#) introduce forward-looking migration with idiosyncratic taste shocks; [Adão et al.](#)

(2019) develop methods for estimating spatial models with rich geography; and Kleinman et al. (2023) add regional capital accumulation. Our model inherits the dynamic structure of Kleinman et al. (2023) and extends it to a multi-sector production environment with input-output linkages—an important extension in the Chinese context, where provinces differ sharply in sectoral composition and a uniform trade-cost reduction therefore has heterogeneous effects across space even before any reallocation occurs.

Production networks and trade. The importance of input-output linkages for the transmission of shocks is established both theoretically and empirically. Acemoglu et al. (2012) show that firm-level shocks can have aggregate effects when the network is sufficiently concentrated; Caliendo and Parro (2015) document that NAFTA welfare gains operated substantially through input-output propagation, with sectors whose intermediate inputs became cheaper passing those cost reductions to downstream users; Antràs and Chor (2013) study how trade costs and contractual frictions shape global value chains; Costinot and Rodríguez-Clare (2014) show that intermediate-input linkages amplify the welfare gains from trade; Baqaee and Farhi (2019) characterize how network structure determines which microeconomic shocks have large aggregate effects; and Boehm et al. (2019) document the propagation of supply-chain disruptions following the 2011 Japan earthquake. In the spatial context, Tombe and Zhu (2019) show that interprovincial trade and migration frictions interact in determining Chinese aggregate productivity, though their framework abstracts from dynamic adjustment and treats capital as exogenous, and Caliendo et al. (2018) show that intermediate-input linkages account for a substantial share of the welfare gains from trade shocks in a spatial model. By embedding a multi-sector production network within a dynamic framework that also features endogenous migration and capital accumulation, this paper addresses how the network amplifies the welfare gains from transport infrastructure, how this amplification evolves over time as factors reallocate, and how it varies across regions depending on their initial sectoral structure.

Transportation infrastructure and economic development. A large empirical literature documents the effects of transport networks on regional outcomes. Quasi-experimental studies of historical and modern rail networks find substantial local gains

from railway access through reduced trade costs and improved market access (Donaldson, 2018; Donaldson and Hornbeck, 2016). Evidence on highways and roads is more mixed: Michaels (2008) documents positive effects on counties directly served by the US Interstate Highway System but negative spillovers to bypassed counties; Duranton et al. (2014) find that rural Indian roads raise agricultural productivity without commensurate non-agricultural gains; Faber (2014) shows that China’s National Trunk Highway System raised market access for peripheral counties but slowed their relative GDP growth; Faber and Gaubert (2019) document ambiguous welfare effects from Mexican tourism infrastructure; and Alder (2025) finds that Ethiopian road improvements raise agricultural productivity with limited structural transformation. A rapidly growing body of work focuses on China’s railway expansion, including Banerjee et al. (2020) on historical transport access and long-run development, Zheng and Kahn (2013) on commuting and megacity labor markets, Lin (2017) on specialization across connected cities, Qin (2017) on urban land capitalization, and Chen and Haynes (2017) on modal substitution and air quality. Relative to these studies, which estimate reduced-form effects on specific local outcomes, this paper provides a unified structural welfare assessment that embeds the reduced-form mechanisms within general equilibrium and quantifies aggregate and distributional implications over a long horizon; the structural approach is also necessary for separating direct effects from spillovers since comparing connected to unconnected regions confounds the benefits of connection with the consequences of competitors becoming more connected.

Regional economics of China. The paper also relates to a broader literature on China’s spatial development, which documents substantial cross-provincial heterogeneity in income, productivity, and market access. Young (2000) documents the fragmentation of China’s internal goods markets due to local protectionism; Bai et al. (2006) estimate substantial cross-provincial variation in the return to capital; Hsieh and Klenow (2009) quantify the productivity costs of factor misallocation; Brandt et al. (2013) attribute persistent labor-productivity gaps to migration restrictions; Tombe (2015) documents sizeable remaining interprovincial trade costs; Hsieh and Moretti (2019) estimate that relaxing migration restrictions could raise GDP by 20–30 percent; Tombe and Zhu (2019) show that

the gains from reducing trade and migration frictions are complementary; and [Tian \(2024\)](#) documents the role of infrastructure in facilitating internal migration. A recurring theme is that trade frictions, migration frictions, and incomplete capital mobility together produce persistent spatial misallocation. Railway infrastructure directly targets the cost of moving goods, but its welfare consequences depend on how it interacts with the other frictions. Our structural decomposition provides an explicit quantification: welfare gains from railway construction are substantially amplified by migration and by capital reallocation, and the production network contributes a further non-trivial share. This decomposition also speaks to an older question in the Chinese development literature about the relative importance of infrastructure, factor mobility, and industrial policy in explaining regional convergence over the reform period ([Fleisher et al., 2010](#); [Fan et al., 2023](#)).

The remainder of the paper is organized as follows. Section 2 introduces the institutional background of China’s railway expansion and describes the data, including the construction of bilateral railway connectivity measures and regional exposure indices. Section 3 develops the theoretical model. Section 4 describes the quantification strategy, including the recovery of model fundamentals, the solution of the dynamic equilibrium, and a set of reduced-form validation exercises that test whether the model’s main mechanisms are present in the data. Section 5 presents the counterfactual results, including aggregate welfare gains, regional and sectoral heterogeneity, a decomposition of the gains into price, wage, migration, production-network, and capital-accumulation channels, and robustness checks. Section 6 concludes.

2 Institutional Background and Data

This section describes the institutional background of China’s railway expansion and introduces the main datasets. The core empirical variation comes from a prefecture-pair panel of railway shortest distances covering 2002–2016. This dataset, combined with multi-regional input-output tables, population census data, and provincial economic statistics, provides the information needed to calibrate the model, recover its fundamentals, and conduct counterfactual experiments.

2.1 China’s Railway Expansion, 2002–2016

China’s railway system underwent a dramatic transformation over our sample period. At the beginning of 2002, the national network comprised approximately 72,000 kilometers of track, serving primarily as a freight and long-distance passenger system with limited intercity passenger service. The network was concentrated in the eastern coastal provinces and the northeast industrial belt, with large areas of central and western China still poorly served despite the upgrading efforts of the late 1990s.

Over the following fifteen years, the Chinese government progressively scaled up railway investment through three overlapping planning cycles: the Tenth Five-Year Plan (2001–2005), the Eleventh Five-Year Plan (2006–2010), and the Twelfth Five-Year Plan (2011–2015). The program had several distinct phases. During the early years (2002–2004), the focus was on continuing to upgrade existing conventional trunk lines and extending the network to underserved regions, with particular attention to east–west corridors linking coastal provinces to the interior. The Medium- and Long-Term Railway Network Plan, first announced in 2004 and revised in 2008, laid out a comprehensive blueprint for subsequent expansion, envisioning a system of “four vertical and four horizontal” passenger-dedicated lines supplemented by extensive regional networks.

The period from 2005 to 2010 saw the beginning of high-speed rail construction. The first dedicated high-speed line (the Beijing–Tianjin intercity railway) opened in 2008 and the longer Wuhan–Guangzhou line entered service in 2009. Construction accelerated markedly after the 2008 financial crisis, as railway investment became a central component of China’s fiscal stimulus package: annual railway fixed-asset investment roughly doubled between 2008 and 2010. By 2016, the national network had grown to over 124,000 kilometers, of which more than 22,000 kilometers were high-speed lines operating at 250 km/h or above. The high-speed network alone carried over 1.4 billion passenger trips in 2016, fundamentally altering intercity travel patterns and, through its effects on passenger services, freeing conventional rail capacity for freight transport.

Three features of this expansion make our sample period particularly well-suited for studying the economic effects of railway construction. First, the scale of investment was

enormous, generating substantial cross-sectional variation in the degree to which different prefecture pairs experienced improvements in railway connectivity. Some inland prefectures that were previously accessible only after lengthy detours became directly connected to the coastal economic centers, while other prefectures experienced more modest improvements. Second, the sample period is long enough—fifteen years—to observe the dynamic adjustment of migration and capital accumulation, which the model predicts to be gradual. Third, the period corresponds to an era of deepening domestic market integration in China, during which railways played a central role in connecting production centers, consumer markets, and labor supply regions.

2.2 Railway Shortest-Distance Data

The main dataset is a prefecture-pair-year panel of railway shortest distances from 2002 to 2016. For each year t , the data record the shortest railway-network distance $RailDist_{ijt}$ between each pair of prefecture-level administrative units i and j . This distance is network-based: it measures the length of the shortest path through the railway network connecting the two prefectures, not the Euclidean distance between them. The crucial feature of this variable is that it changes over time as new railway lines are built and existing connections are upgraded. Geographic distance between two prefectures is fixed; railway network distance is not.

The dataset has a bilateral panel structure $\{RailDist_{ijt}\}$ for $i, j = 1, \dots, N$ and $t = 2002, \dots, 2016$, where N is the number of prefecture-level units. Railway distance is symmetric ($RailDist_{ijt} = RailDist_{jit}$) and normalized so that $RailDist_{iit} = 0$. To measure year-on-year improvements in railway access, we define bilateral railway improvement as the negative log change in railway shortest distance:

$$RailImprove_{ijt} \equiv -\Delta \ln RailDist_{ijt} = \ln RailDist_{ij,t-1} - \ln RailDist_{ijt}. \quad (1)$$

A positive value indicates that the shortest railway path between prefectures i and j became shorter. For cumulative improvements over the full sample, we define $RailImprove_{ij}^{02-16} = \ln RailDist_{ij,2002} - \ln RailDist_{ij,2016}$.

To summarize how changes in the railway network affect each region’s overall connectivity, we construct a regional railway exposure measure:

$$RailExposure_{it} = - \sum_{j \neq i} \omega_{ij0} (\ln RailDist_{ijt} - \ln RailDist_{ij,2002}), \quad (2)$$

where ω_{ij0} is a predetermined bilateral weight fixed at its 2002 value. Our baseline uses initial trade-share weights $\omega_{ij0} = \sum_h \psi^h S_{ij0}^h / \sum_{m \neq i} \sum_h \psi^h S_{im0}^h$, which give greater weight to railway improvements with initially important trading partners. We also report results using population weights, GDP weights, and equal weights.

The railway distance data serve two distinct purposes in our analysis. First, they provide direct measures of bilateral connectivity changes for the reduced-form empirical exercises. Second, they are used to construct the railway-induced component of trade-cost changes for the structural counterfactual, as we describe in Section 4.

2.3 Additional Data Sources

The quantitative analysis requires several additional datasets, which we summarize in Table 1 and describe briefly here.

Interprovincial trade flows. We use China’s multi-regional input-output (MRIO) tables for the benchmark years 2002, 2007, 2012, and 2017. The 2002 table coincides with the first year of our sample, and the 2017 table is used to pin down the late-sample trade structure. Each table covers thirty provinces and forty-two sectors, which we aggregate into three broad categories: agriculture, manufacturing, and services. The sector mapping follows the original IO classification for 2002 and 2007, with minor reclassification for 2012 and 2017 to accommodate revisions in the official industry codes. The MRIO tables record the value of goods shipped from each origin province-sector to each destination province-sector, including both intermediate use and final demand. From these flows, we construct bilateral trade shares for each sector, province pair, and benchmark year. Annual trade shares between benchmark years are obtained by piecewise cubic Hermite interpolation of the underlying trade-cost levels, followed by Gaussian smoothing to remove high-frequency noise.

Migration flows. Interprovincial migration probabilities are constructed from three sources of population data: the 2005 one-percent population survey, the 2010 population census, and the 2015 one-percent population survey. Each source records the province of current residence and the province of residence five years earlier, from which we construct five-year origin-destination migration matrices. We convert these into annual transition matrices using a matrix root procedure based on real Schur decomposition, with corrections to ensure non-negativity and row normalization. Because the migration data do not distinguish sectors, we assume that the sector-specific migration matrix equals the aggregate matrix for all sectors. Annual migration probabilities for non-census years are obtained by linear interpolation of the log bilateral migration cost between adjacent census years; for the early-sample years that precede the first census observation, we extrapolate backward using the slope implied by the first two census points.

Employment by province and sector. Provincial employment by sector (agriculture, manufacturing, and services) is obtained from provincial statistical yearbooks and the *China Population and Employment Statistics Yearbook*, covering the full sample period. These data determine the sectoral labor allocation, which is the key labor state variable in the model.

Provincial capital stocks. We construct provincial capital stocks using the perpetual inventory method applied to real fixed-asset investment, following [Bai et al. \(2006\)](#). Capital growth rates by province are used to initialize the model’s capital accumulation block.

Provincial GDP by sector. Real GDP by province and sector, obtained from provincial statistical yearbooks, is used to calibrate productivity growth and to validate the model’s GDP predictions.

National input-output table. The national IO table is used to calibrate the production function and consumption share parameters. We use the 2002 table as the baseline—consistent with our sample starting date—and verify stability across benchmark years.

Amenity indicators. To decompose local amenities, we collect provincial data on air quality (share of days with good or excellent air quality), medical resources (physicians per thousand population), education resources (college enrollment relative to population),

transport density (highway kilometers per square kilometer), climate comfort (the absolute deviation of annual mean temperature from a human-comfort benchmark, entered with a negative sign), and public safety (per capita criminal cases, entered with a negative sign).

Table 1. Summary of Data Sources

Data	Source	Coverage
Railway shortest distance	Railway network GIS	2002–2016
Interprovincial trade flows	MRIO tables	2002, 07, 12, 17
Migration flows	Census & 1% surveys	2005, 2010, 2015
Employment by sector	Provincial yearbooks	2002–2017
Capital stock	PIM from investment data	2002–2017
GDP by sector	Provincial yearbooks	2002–2017
National IO table	NBS	2002
Amenity indicators	Various	2002–2017

3 Model

This section develops the dynamic spatial general equilibrium model. Time is discrete, indexed by $t = 0, 1, 2, \dots$. The economy consists of N regions (provinces) indexed by i, j, m and J sectors indexed by h, o, s . Each region produces goods in every sector. Within each sector, goods are differentiated by region of origin and traded across regions subject to iceberg trade costs. Workers are mobile across regions within their sector but cannot switch sectors. Capital is immobile across regions but freely allocated across sectors within a region.

3.1 Production and Trade

Region i sector h produces gross output y_{it}^h using labor l_{it}^h , capital k_{it}^h , and intermediate inputs $\{x_{it}^{h,o}\}_{o=1}^J$ with the Cobb–Douglas production function:

$$y_{it}^h = z_{it}^h \left[\left(\frac{l_{it}^h}{\mu^h \gamma^h} \right)^{\mu^h} \left(\frac{k_{it}^h}{(1 - \mu^h) \gamma^h} \right)^{1 - \mu^h} \right]^{\gamma^h} \prod_{o=1}^J \left(\frac{x_{it}^{h,o}}{\gamma^{h,o}} \right)^{\gamma^{h,o}}, \quad (3)$$

where z_{it}^h is total factor productivity, $\gamma^h \in (0, 1]$ is the value-added share in gross output, $\mu^h \in (0, 1)$ is the labor share in value added, and $\gamma^{h,o} \geq 0$ is the cost share of sector- o intermediates used by sector h , satisfying $\sum_{o=1}^J \gamma^{h,o} = 1 - \gamma^h$. The normalization

constants in the denominators ensure that, under cost minimization, expenditure shares on each input equal the corresponding exponents.

Firms are perfectly competitive. Capital is freely mobile across sectors within each region, yielding a common rental rate r_{it} . Labor is sector-specific, so wages w_{it}^h may differ across sectors within the same region. Cost minimization implies that factor payments are proportional to revenue:

$$w_{it}^h l_{it}^h = \mu^h \gamma^h p_{it}^h y_{it}^h, \quad (4)$$

$$r_{it} k_{it}^h = (1 - \mu^h) \gamma^h p_{it}^h y_{it}^h, \quad (5)$$

$$p_{it}^o x_{it}^{h,o} = \gamma^{h,o} p_{it}^h y_{it}^h, \quad (6)$$

where p_{it}^h is the factory-gate price of sector- h output in region i . Dividing (5) by (4) gives the capital-labor payment relation $r_{it} k_{it}^h = [(1 - \mu^h)/\mu^h] w_{it}^h l_{it}^h$, which links capital demand in each sector to the wage bill. Substituting the optimal input demands back into the production function yields the factory-gate price as unit cost divided by productivity:

$$p_{it}^h = \frac{C_{it}^h}{z_{it}^h}, \quad C_{it}^h \equiv \left[(w_{it}^h)^{\mu^h} r_{it}^{1-\mu^h} \right]^{\gamma^h} \prod_{o=1}^J (p_{it}^o)^{\gamma^{h,o}}, \quad (7)$$

where C_{it}^h is the unit cost index. The input-output structure enters through the product term $\prod_o (p_{it}^o)^{\gamma^{h,o}}$: a decline in the price of sector- o goods reduces the marginal cost of every sector that uses sector o as an input. This is the channel through which railway-induced reductions in goods prices propagate across sectors.

Shipping sector- h goods from region j to region i incurs an iceberg trade cost $\tau_{ijt}^h \geq 1$, with $\tau_{iit}^h = 1$ (no cost for local trade) and $\tau_{ijt}^h = \tau_{jit}^h$ (symmetry). To deliver one unit of the good, τ_{ijt}^h units must be shipped from the origin, with $\tau_{ijt}^h - 1$ units “melting” in transit. The delivered price is therefore $p_{ijt}^h = \tau_{ijt}^h C_{jt}^h / z_{jt}^h$, where the cost structure depends on the source region j .

Consumers in each region aggregate sector- h varieties from different origins using a CES aggregator with elasticity of substitution $1 + \theta$ ($\theta > 0$). Expenditure minimization yields the sectoral price index and the bilateral expenditure share:

$$p_{it}^h = \left[\sum_{j=1}^N (p_{ijt}^h)^{-\theta} \right]^{-1/\theta}, \quad S_{ijt}^h = \left(\frac{p_{ijt}^h}{p_{it}^h} \right)^{-\theta}, \quad (8)$$

where S_{ijt}^h is the share of region i 's sector- h expenditure on goods from region j , and $\sum_j S_{ijt}^h = 1$. A reduction in the bilateral trade cost τ_{ijt}^h lowers the delivered price p_{ijt}^h , increases the expenditure share S_{ijt}^h , and reduces the sectoral price index p_{it}^h by expanding the set of cheaply accessible varieties.

Across sectors, final consumption follows a Cobb–Douglas aggregator with expenditure shares $\psi^h > 0$ summing to one:

$$c_{it} = \prod_{h=1}^J \left(\frac{c_{it}^h}{\psi^h} \right)^{\psi^h}, \quad p_{it} = \prod_{h=1}^J (p_{it}^h)^{\psi^h}. \quad (9)$$

The aggregate price index p_{it} is the cost-of-living index faced by consumers in region i .

3.2 Households, Migration, and Lifetime Welfare

A sector- h worker in region i at time t receives instantaneous indirect utility $\tilde{u}_{it}^h = b_{it} w_{it}^h / p_{it}$, where $b_{it} > 0$ is a local amenity capturing quality of life, public services, climate, and other non-pecuniary attributes. Utility increases with the real wage w_{it}^h / p_{it} and with local amenity b_{it} .

At the end of each period, each worker chooses where to live in the next period. Workers cannot switch sectors: a manufacturing worker remains in manufacturing regardless of which region she moves to. Moving from region i to region j requires paying a migration cost $\kappa_{ijt} \geq 1$, with $\kappa_{iit} = 1$ (staying is free) and $\kappa_{ijt} = \kappa_{jit}$ (symmetry). The dynamic location-choice problem is:

$$v_{it}^h = \ln \tilde{u}_{it}^h + \max_{j \in \{1, \dots, N\}} \left\{ \beta E_t E_\varepsilon [v_{j,t+1}^h] - \ln \kappa_{ijt} + \rho \varepsilon_{jt} \right\}, \quad (10)$$

where $\beta \in (0, 1)$ is the discount factor, ε_{jt} is an i.i.d. standard Gumbel taste shock, and $\rho > 0$ governs the dispersion of idiosyncratic location preferences. A larger ρ implies that migration choices are less sensitive to systematic differences in utility across destinations.

Equation (10) is the structural foundation of labor mobility in the model. It nests the standard static gravity model of migration as a special case (when $\beta = 0$) and adds the crucial forward-looking component: workers value not only current real income but also the *option value* of future mobility. A region that currently offers moderate wages but provides access to many high-value destinations (e.g., because it is centrally located and

has low migration costs to many places) may be more attractive than a high-wage region that is isolated.

Define the expected lifetime value $V_{it}^h \equiv E_\varepsilon[v_{it}^h]$ and the dynamic welfare auxiliary $u_{it}^h \equiv \exp(\beta V_{it}^h / \rho)$. By the standard Gumbel log-sum formula, the expected lifetime value satisfies:

$$V_{it}^h = \ln b_{it} + \ln \frac{w_{it}^h}{p_{it}} + \rho \ln \sum_{j=1}^N \frac{u_{j,t+1}^h}{\kappa_{ijt}^{1/\rho}}, \quad (11)$$

and the welfare auxiliary obeys the recursion:

$$u_{it}^h = \left[b_{it} \frac{w_{it}^h}{p_{it}} \left(\sum_{j=1}^N \frac{u_{j,t+1}^h}{\kappa_{ijt}^{1/\rho}} \right)^\rho \right]^{\beta/\rho}. \quad (12)$$

The welfare recursion (12) is the key welfare object in our analysis. It links current welfare to current amenity-adjusted real income (the first term in brackets) and to the discounted option value of all future migration opportunities (the second term). Railway construction affects both: it raises current real income by lowering consumer prices and potentially increasing wages, and it improves the option value by altering the spatial distribution of future welfare across destinations.

The Gumbel distributional assumption yields closed-form migration probabilities:

$$D_{ijt}^h = \frac{u_{j,t+1}^h / \kappa_{ijt}^{1/\rho}}{\sum_{m=1}^N u_{m,t+1}^h / \kappa_{imt}^{1/\rho}}. \quad (13)$$

The probability of moving from i to j is higher when destination j offers higher future lifetime welfare ($u_{j,t+1}^h$) and when the migration cost is lower (κ_{ijt}). Given these probabilities, sectoral labor in each region evolves according to:

$$l_{j,t+1}^h = \sum_{i=1}^N D_{ijt}^h l_{it}^h. \quad (14)$$

Because workers cannot switch sectors, total employment within each sector is conserved:

$$\sum_j l_{j,t+1}^h = \sum_i l_{it}^h.$$

3.3 Capital Accumulation

Capital in each region is owned by immobile investors who solve a standard CRRA consumption–investment problem:

$$\max_{\{c_{i,t+s}^k, k_{i,t+s+1}\}_{s \geq 0}} E_t \sum_{s=0}^{\infty} \beta^s \frac{(c_{i,t+s}^k)^{1-1/\psi}}{1-1/\psi}, \quad (15)$$

subject to the budget constraint $r_{it}k_{it} = p_{it}[c_{it}^k + k_{i,t+1} - (1-\delta)k_{it}]$, where δ is the depreciation rate and $\psi > 0$ is the intertemporal elasticity of substitution. Defining the gross investment return $R_{it} \equiv 1 - \delta + r_{it}/p_{it}$ and the consumption share $\varsigma_{it} \equiv c_{it}^k/(R_{it}k_{it})$, the capital law of motion becomes $k_{i,t+1} = (1-\varsigma_{it})R_{it}k_{it}$, and the Euler equation under perfect foresight simplifies to:

$$\varsigma_{it}^{-1} = 1 + \beta^\psi \varsigma_{i,t+1}^{-1} R_{i,t+1}^{\psi-1}. \quad (16)$$

Capital accumulation is the third dynamic margin in the model, alongside labor migration and the evolution of trade shares. Railway construction can stimulate investment by raising capital returns: improved market access increases demand for locally produced goods, which raises the marginal product of capital and the investment return R_{it} . Over time, higher investment increases the capital stock, which expands productive capacity, lowers marginal costs, and affects wages and prices throughout the economy.

3.4 Market Clearing and Dynamic Equilibrium

Within each region, capital market clearing requires that total capital income equals the sum of sectoral capital payments:

$$r_{it}k_{it} = \sum_{h=1}^J \frac{1-\mu^h}{\mu^h} W_{it}^h, \quad W_{it}^h \equiv w_{it}^h t_{it}^h. \quad (17)$$

Goods-market clearing requires that each source region-sector's revenue equals total demand from all destinations. Total expenditure in destination j on sector- h goods combines final consumption demand and intermediate input demand from all producing sectors:

$$E_{jt}^h = \sum_{o=1}^J \frac{\psi^h \gamma^o + \gamma^{o,h}}{\mu^o \gamma^o} W_{jt}^o. \quad (18)$$

The goods-market clearing condition for source region i , sector h , is then:

$$\frac{W_{it}^h}{\mu^h \gamma^h} = \sum_{j=1}^N S_{jit}^h E_{jt}^h, \quad (19)$$

where S_{jit}^h denotes the share of destination j 's sector- h expenditure on goods from source i .

Definition 1 (Dynamic Equilibrium). *Given initial conditions $\{l_{i0}^h, k_{i0}, k_{i1}, S_{ij0}^h, D_{ij0}^h\}$ and sequences of exogenous fundamentals $\{z_{it}^h, b_{it}, \tau_{ijt}^h, \kappa_{ijt}\}_{t \geq 0}$, a dynamic equilibrium is a sequence of allocations and prices $\{l_{it}^h, k_{it}, w_{it}^h, r_{it}, p_{it}^h, p_{it}, S_{ijt}^h, D_{ijt}^h, u_{it}^h, R_{it}, \varsigma_{it}\}_{t \geq 0}$ such that:*

- (i) *firms minimize costs and price equals marginal cost;*
- (ii) *consumers minimize expenditure within and across sectors;*
- (iii) *workers solve the dynamic location-choice problem;*
- (iv) *investors solve the consumption–investment problem;*
- (v) *capital, labor, and goods markets clear in every period.*

The dynamic equilibrium traces a transition path along which labor reallocates across regions through migration, capital accumulates through investment, and prices, wages, and trade shares adjust to clear all markets. Railway construction shifts this path by reducing bilateral trade costs, which triggers a cascade of general equilibrium responses: lower delivered prices, changed trade patterns, altered production costs through input-output linkages, modified wages and capital returns, shifted migration incentives, and ultimately changed lifetime welfare for workers in every region and sector.

3.5 Welfare Measurement

The model provides a natural welfare measure through the dynamic value function. Because u_{it}^h is a monotone transformation of expected lifetime utility V_{it}^h , welfare comparisons based on u_{it}^h are equivalent to comparisons based on lifetime utility. The consumption-equivalent variation (CEV) for sector- h workers in region i between a baseline (BL) and counterfactual (CF) economy is:

$$CEV_{it}^h = \frac{u_{it}^{h,BL}}{u_{it}^{h,CF}} - 1. \quad (20)$$

A positive CEV_{it}^h means that workers are better off in the baseline than in the counterfactual. Aggregate welfare is the employment-weighted average:

$$CEV_t = \sum_{i=1}^N \sum_{h=1}^J \frac{l_{it}^h}{\sum_{m,s} l_{mt}^s} CEV_{it}^h. \quad (21)$$

This welfare measure is comprehensive: it incorporates not only current real income and amenities but also the discounted value of all future migration options. It therefore captures the dynamic welfare effects of infrastructure investment, including the gains from future labor reallocation that have not yet occurred.

4 Quantification

This section describes how we bring the model to the data. The goal is to construct a fully specified baseline economy that matches observed Chinese provincial data in 2002 and to simulate its dynamic path under alternative assumptions about railway connectivity. The analysis proceeds in four stages. We first set structural parameters. We then recover the model’s unobserved fundamentals—productivity, trade costs, migration costs, and amenities—from the observed data. With these in hand, we construct the initial equilibrium in 2002 and solve the dynamic transition forward in time. Finally, we describe how railway data are used to construct the counterfactual trade-cost path. Throughout, our priority is to explain the logic of the procedure; the technical derivations and algorithmic details are collected in Appendix.

4.1 Overview and Identification Strategy

The central challenge in quantifying a spatial model with rich heterogeneity is that many of the objects of interest—sectoral productivity, bilateral trade costs, local amenities, and lifetime welfare—are not directly observed and cannot in general be recovered from any single data source. Our calibration strategy addresses this challenge through two design choices.

First, we exploit the *structural restrictions of the model itself* to back out unobservables from observable cross-sectional patterns. Bilateral trade costs leave a signature in the matrix of trade shares; migration costs leave a signature in the matrix of migration

probabilities; productivity leaves a signature in the home-trade share combined with factor prices. Each of these signatures has a closed-form inversion that does not require estimating additional parameters.

Second, we confront a fundamental *identification problem*: cross-sectional share data can only identify *relative* levels of productivity, welfare, and amenity across regions, not their common national trend. If all provinces became twice as productive, all trade shares would remain unchanged. To preserve the economic content of aggregate trends—which are plainly visible in China’s growth experience—we adopt a *trend–dispersion decomposition*: each unobserved fundamental is written as the product of a national common component (pinned down by aggregate time-series data) and a relative regional component (pinned down by cross-sectional shares). This decomposition ensures that the calibrated model reproduces both the aggregate growth path of the Chinese economy and the cross-sectional distribution of activity across provinces.

The calibration is sequential. Each quantity is recovered using only information that has been determined in an earlier step or observed directly in the data. This sequencing avoids circularity and makes the procedure transparent.

4.2 Structural Parameters

The model’s structural parameters fall into two groups. The first group consists of production and consumption parameters—value-added shares, labor shares, intermediate-input coefficients, and final consumption shares—which are computed directly from the 2002 national input–output table after aggregating the original forty-two sectors into three broad categories (agriculture, manufacturing, and services). These parameters govern the sector-level technology and the input–output network through which railway-induced cost reductions propagate. The calibrated values are reported in Table 2.

The second group consists of parameters that cannot be read directly from the input–output table: the trade elasticity, the migration dispersion parameter, the discount factor, the depreciation rate, and the intertemporal elasticity of substitution. For these we rely on values that are standard in the quantitative spatial and macroeconomic literatures, with

attention to the Chinese context. The trade elasticity is set to a value consistent with estimates from the international trade literature; the migration dispersion parameter is set to reflect the substantial frictions imposed by China’s household registration (*hukou*) system; and the remaining macroeconomic parameters follow conventional choices for an annual calibration. Table 3 summarizes these values and their sources.

Table 2. Calibrated Production and Consumption Parameters

	Agriculture	Manufacturing	Services
Value-added share	0.650	0.340	0.600
Labor share in value added	0.851	0.340	0.436
Final consumption share	0.095	0.446	0.459
<i>Intermediate input intensity (row sector uses column sector)</i>			
Agriculture	0.120	0.170	0.060
Manufacturing	0.035	0.530	0.095
Services	0.010	0.210	0.180

Notes: Computed from the 2002 national input–output table after aggregation to three sectors. Each row of the intermediate input matrix sums to one minus the value-added share of the row sector.

Table 3. Other Structural Parameters

Parameter	Description	Value	Source
θ	Trade elasticity	5.0	Quantitative trade literature
ρ	Migration dispersion	5.8	Consistent with <i>hukou</i> frictions
β	Annual discount factor	0.95	Standard
δ	Capital depreciation rate	0.08	Bai et al. (2006)
ψ	Intertemporal elasticity of substitution	1.5	Macroeconomic literature

4.3 Recovering Model Fundamentals

With structural parameters in hand, we turn to the model’s unobserved fundamentals. The recovery proceeds in six conceptual blocks, ordered so that each block uses only previously determined quantities.

We begin with bilateral trade costs, which are identified by comparing the share of region i ’s expenditure on region j ’s goods with the share of j ’s expenditure on i ’s goods, normalized by the two regions’ own-purchase shares. The key insight—due to [Head and Ries \(2001\)](#)—is that combining these four shares in a specific ratio eliminates all unob-

served cost and productivity terms, leaving only the bilateral trade cost. We apply this inversion to each of the four benchmark multi-regional input–output tables (2002, 2007, 2012, 2017), obtaining bilateral trade-cost levels for each sector and benchmark year. Annual values between benchmarks are obtained by smooth interpolation, and values beyond 2017 are held at their late-sample level.

The same logic applies to migration costs. Observed bilateral migration probabilities are generated by a discrete-choice problem in which workers compare destinations by their lifetime welfare net of migration costs. Combining four migration probabilities— i -to- j , j -to- i , i -to- i , and j -to- j —in an analogous ratio eliminates the welfare terms and isolates the bilateral migration cost. We construct annual migration matrices from the 2005, 2010, and 2015 census data using a matrix-root procedure described in Appendix and invert them period by period.

Given trade costs and observed trade shares, the goods-market clearing condition determines the vector of sectoral labor incomes across provinces. Intuitively, this condition requires that revenue earned by each province-sector equals the value of demand for its output, summed across all destinations and all downstream sectors. The condition takes the form of a fixed-point problem whose solution identifies relative labor incomes up to a scalar. We pin down the scale by requiring that the total wage bill in the model equals the national labor compensation reported in the official flow-of-funds accounts. Wages are then recovered by dividing labor income by observed sectoral employment. Given wage bills and observed provincial capital stocks (constructed using the perpetual inventory method), the capital-market clearing condition determines the rental rate of capital in each province without further estimation.

Productivity is the next object to recover, and it is decomposed into a national sectoral trend and a relative regional component. The regional component is recovered from the pattern of home-trade shares together with factor prices: intuitively, a province that sources most of its demand locally despite facing high unit costs must have correspondingly high productivity. Because productivity and prices appear jointly in the relevant equations, the recovery proceeds by a nested iteration. The national trend is disciplined

separately using sector-level Solow residuals constructed from national accounts, with an adjustment factor that maps value-added productivity growth to the gross-output productivity that enters the model. The two components are multiplied to obtain the full productivity series.

Local amenities are recovered last, by inverting the welfare equation: given wages, prices, migration costs, and the recovered welfare values, the residual that rationalizes observed location choices is interpretable as the amenity. This residual, however, confounds a national common trend (e.g., rising average environmental quality) with province-specific deviations. We separate the two using a reduced-form projection onto observable quality-of-life indicators—air quality, medical and educational resources, transport density, climate, and public safety—estimated with province fixed effects. The projected national trend captures the common component, while the province-specific residual and fixed effects capture the spatial deviation. Amenities therefore receive an empirical content that is grounded in observable quality-of-life measures rather than being defined purely as an unobserved residual.

The final step of calibration converts the recovered level sequences into the change rates that feed into the dynamic solution. For each fundamental, we compute the year-on-year ratio of its calibrated level. Beyond the data window (after 2017), all change rates are set to one, reflecting the assumption that the economy converges to a stationary equilibrium in the long run.

4.4 Dynamic Solution

The calibrated objects collectively define the initial equilibrium of the model at $t = 0$, corresponding to the year 2002. At this date the model has all the information needed to compute current payoffs and to initialize the dynamic laws of motion: observed sectoral employment and provincial capital stocks; observed trade and migration matrices; calibrated wages and capital rental rates; calibrated sectoral and aggregate price indices; and an initial investment return calibrated to match the observed capital growth rate between 2002 and 2003. The initial consumption–investment share of households is set slightly

below its long-run steady-state value, consistent with the high household saving rates that characterized the Chinese economy in the early 2000s. This choice primarily affects the speed of transition, not the long-run path.

From this initial condition, we solve the dynamic equilibrium using *exact-hat algebra*, a method that exploits the model’s structure to express the equilibrium conditions in terms of changes in variables rather than their levels (Dekle et al., 2008; Caliendo and Parro, 2015). The key advantage is that the dynamic path of the economy can be computed using only the initial equilibrium and the change rates of exogenous fundamentals. The absolute levels of unobservable fundamentals—which are not identified from the data—never need to be estimated.

Intuitively, the solution alternates between two passes. In the *forward pass*, given a conjectured path of future welfare, the solver steps through time: workers migrate based on their expectations of future welfare; labor allocations update; investors set the rate of capital accumulation; and within each period, goods markets clear by determining wages, prices, and capital returns consistent with the updated factor supplies and the exogenous changes in productivity and trade costs. In the *backward pass*, the welfare recursion and the Euler equation for saving are solved from a terminal steady-state condition backward to the initial period, generating an updated path of future welfare. The two passes are iterated until they agree, at which point the entire dynamic path is mutually consistent: expectations are rational, all markets clear in every period, and the economy converges to a stationary equilibrium in the long run. Appendix D provides the complete system of equations in changes; the algorithmic details, including convergence tolerances, damping, and terminal-horizon choices, are documented in the replication code.

We solve the economy over a fifty-year horizon starting in 2002, which is long enough for the endogenous state variables (labor allocations, capital stocks, welfare) to approach their new long-run values under the calibrated exogenous path. The data period (2002–2017) is used for calibration and model validation; the remaining years trace the dynamic adjustment implied by the calibrated fundamentals and provide the horizon over which long-run welfare effects are evaluated.

4.5 Counterfactual Design

The core quantitative exercise compares the baseline economy—in which all observed fundamentals, including railway-induced trade-cost changes, follow their actual 2002–2017 paths—with a counterfactual economy in which railway distances are held fixed at their 2002 levels while all other fundamentals evolve as observed. The welfare difference between the two economies isolates the contribution of railway construction.

The empirical challenge is that observed changes in trade costs reflect not only railway expansion but also highway investment, port development, regulatory liberalization, and other forces. Attributing all trade-cost change to railways would overstate the railway effect; ignoring all of it would understate it. We resolve this with a two-step procedure that combines a reduced-form estimate of the railway–trade-cost relationship with a structural substitution into the counterfactual path.

The first step estimates the elasticity of calibrated trade costs with respect to bilateral railway connectivity. Using the panel of recovered trade costs τ_{ijt}^h for each sector h , province pair (i, j) , and year t , we estimate:

$$\ln \tau_{ijt}^h = \eta^h \ln \text{RailDist}_{ijt} + \alpha_{ij}^h + \lambda_t^h + X'_{ijt} \Gamma^h + \varepsilon_{ijt}^h, \quad (22)$$

where RailDist_{ijt} is the railway shortest-path distance between prefectures in provinces i and j in year t ; α_{ij}^h is a province-pair fixed effect that absorbs all time-invariant bilateral characteristics (geographic distance, terrain, cultural ties, historical administrative boundaries); λ_t^h is a year fixed effect that absorbs common time shocks and nationwide policy changes; and X_{ijt} is a vector of time-varying bilateral controls (highway connectivity, port access, and whether the pair spans a provincial administrative border that changed status during the sample). The coefficient of interest η^h is identified from *within-pair* changes in railway connectivity over time, purged of any confounding factors that are either time-invariant at the pair level or nationwide in scope. The model predicts $\eta^h > 0$: a reduction in railway distance lowers trade costs.

We estimate (22) separately by sector and also pooled across sectors. As a robustness check, we split railway distance into its conventional and high-speed components, since the

two may have different elasticities for goods transport. We also consider specifications that use a measure of *effective* railway time rather than distance, to capture the fact that high-speed lines primarily serve passengers and affect goods movement only indirectly through reallocation of freight capacity on conventional lines. The reduced-form results serve a dual purpose: they validate the model’s prediction that railway connectivity reduces trade costs, and they provide the elasticity $\hat{\eta}^h$ used in the counterfactual construction.

The second step constructs the counterfactual trade-cost path by removing the component of observed trade-cost changes that is attributable to railway construction. In levels:

$$\tau_{ijt}^{h,CF} = \tau_{ijt}^{h,BL} \left(\frac{RailDist_{ij,2002}}{RailDist_{ijt}} \right)^{\hat{\eta}^h}, \quad (23)$$

where $\tau_{ijt}^{h,BL}$ is the baseline calibrated trade cost and $RailDist_{ij,2002}$ is the bilateral railway distance observed in 2002. The ratio $RailDist_{ij,2002}/RailDist_{ijt}$ captures the cumulative railway-induced improvement in connectivity between the two provinces relative to the 2002 baseline; raising this ratio to the power $\hat{\eta}^h$ converts the connectivity change into an equivalent trade-cost change. Because railway construction generally reduces $RailDist_{ijt}$ over time, the ratio exceeds one and $\tau_{ijt}^{h,CF}$ exceeds $\tau_{ijt}^{h,BL}$: trade costs in the counterfactual are higher, reflecting the absence of railway-induced improvements. Equivalently, in change rates, the counterfactual satisfies

$$\dot{\tau}_{ij,t+1}^{h,CF} = \dot{\tau}_{ij,t+1}^{h,BL} \cdot \left(\frac{RailDist_{ij,t+1}}{RailDist_{ijt}} \right)^{\hat{\eta}^h}, \quad (24)$$

which strips the year-on-year railway-induced component out of the observed trade-cost changes while preserving all non-railway components.

This construction has two desirable properties. First, it is internally consistent: the same structural trade cost that enters the model’s equilibrium is the object whose elasticity we estimate, so the reduced-form and structural exercises speak the same language. Second, the pair fixed effects in (22) ensure that the counterfactual is identified from temporal variation in railway connectivity rather than from cross-sectional correlations between railway placement and trade costs. This distinction matters because railway routes are not randomly assigned—lines are more likely to be built between economi-

cally important origin-destination pairs—so a cross-sectional regression would confound the causal effect of railways on trade costs with the selection of railway routes onto high-trade corridors. Within-pair variation is immune to this concern as long as the timing of improvements is not itself responsive to unobserved, time-varying trade frictions; we discuss this identification assumption and provide supporting evidence later.

All exogenous fundamentals other than trade costs are held at their baseline paths in the counterfactual:

$$\dot{z}_{it}^{h,CF} = \dot{z}_{it}^{h,BL}, \quad \dot{b}_{it}^{CF} = \dot{b}_{it}^{BL}, \quad \dot{\kappa}_{ijt}^{CF} = \dot{\kappa}_{ijt}^{BL}. \quad (25)$$

Productivity, amenities, and migration costs therefore evolve identically in the two economies; the only source of divergence is the railway-induced component of trade costs. Both economies are initialized at the same 2002 equilibrium and solved using the same dynamic algorithm, so any difference in their transition paths—including wages, prices, migration flows, capital stocks, and welfare—is attributable entirely to the railway effect.

The counterfactual welfare effect for workers in sector h and province i at horizon t is measured by the consumption-equivalent variation defined in Section 3.5, and aggregate welfare gains are obtained by employment-weighted averaging. To understand *why* railway construction delivers welfare gains, we supplement the main counterfactual with a series of restricted counterfactuals in which specific channels are shut down one at a time: a no-migration counterfactual that holds the labor allocation fixed at its initial distribution; a fixed-capital counterfactual that holds capital stocks at their baseline paths; and a no-input-output counterfactual that collapses the multi-sector production network into a value-added-only economy. Comparing the welfare gains across these restricted counterfactuals with those in the full model isolates the contribution of labor reallocation, capital accumulation, and network propagation, respectively.

Finally, as a robustness check, we consider two alternative ways of constructing the railway counterfactual. The first replaces the estimated elasticity $\hat{\eta}^h$ with a shift-share attribution that assigns to railways the portion of observed trade-cost changes that is correlated with railway distance changes within each pair. The second considers a partial counterfactual that removes only post-2008 railway construction (corresponding to the

high-speed rail era) while retaining earlier improvements in the conventional network. Both alternatives yield counterfactual welfare effects that are qualitatively similar to the baseline.

5 Results

This section presents the empirical and quantitative results. The analysis is organized around the causal chain developed in Sections 3 and 4. We first provide reduced-form evidence that railway improvements are associated with lower recovered trade costs, larger bilateral trade shares, changes in real income, and migration responses consistent with the model. We then evaluate the model’s in-sample fit during 2002–2017. Having established that the calibrated economy reproduces the main aggregate, sectoral, and spatial patterns in the data, we turn to the central counterfactual exercise: comparing the observed baseline economy with a counterfactual economy in which railway distances are held fixed at their 2002 levels. We report aggregate welfare gains, regional and sectoral heterogeneity, dynamic adjustment, mechanism decompositions, and robustness checks.

The main message is that railway construction generates economically meaningful welfare gains that unfold gradually. The immediate gains operate through lower delivered prices and changes in market access. Over time, these static gains are amplified by production-network propagation, labor reallocation, and regional capital accumulation. The dynamic margins are quantitatively important: a large share of the long-run gains is not visible at short horizons. This timing is central for evaluating infrastructure investment, since conventional reduced-form estimates at short horizons or static general equilibrium calculations capture only part of the relevant welfare effect.

5.1 Reduced-Form Evidence

We begin by documenting the reduced-form relationships that discipline the structural counterfactual. These exercises are not intended to replace the model; rather, they provide direct empirical support for the key mechanisms through which the model interprets railway expansion. The specifications use within-pair and within-region variation generated

by the staggered construction of railway lines. Pair fixed effects absorb time-invariant geography, historical trade links, terrain, and cultural proximity. Year fixed effects absorb national business-cycle fluctuations, macroeconomic policy, and aggregate transport-sector trends.

The first exercise estimates the relationship between railway connectivity and the recovered bilateral trade costs. Using the calibrated trade-cost series τ_{ijt}^h described in Section 4.3, we estimate equation (22). Table 4 reports the results. The coefficient on railway distance is positive in all specifications: shorter railway paths are associated with lower trade costs. The pooled elasticity is 0.236, meaning that a 10 percent reduction in railway shortest-path distance is associated with a 2.36 percent decline in recovered bilateral trade costs. The estimate is largest in manufacturing, consistent with manufacturing's greater dependence on interregional shipments and intermediate-input flows. The coefficient remains stable after adding time-varying controls for highway connectivity and port access, suggesting that the railway measure captures a distinct component of transport-cost change rather than merely proxying for broader infrastructure investment.

Table 4. Railway Connectivity and Recovered Trade Costs

	Dependent variable: $\ln \tau_{ijt}^h$				
	Pooled (1)	Agriculture (2)	Manufacturing (3)	Services (4)	Pooled (5)
$\ln \text{RailDist}_{ijt}$	0.236*** (0.041)	0.181*** (0.052)	0.279*** (0.061)	0.211*** (0.049)	0.224*** (0.044)
Highway connectivity					-0.037** (0.016)
Port access					-0.021* (0.012)
Province-pair fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Sector fixed effects	Yes	No	No	No	Yes
Time-varying controls	No	No	No	No	Yes
Observations	5,220	1,740	1,740	1,740	5,220
R^2	0.982	0.973	0.986	0.979	0.984

Notes: The unit of observation is a province-pair-sector-year. The dependent variable is the log of the recovered bilateral iceberg trade cost. RailDist_{ijt} is the railway shortest-path distance between provinces i and j . Standard errors clustered by province pair are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The second exercise examines whether the same railway improvements are reflected in bilateral trade shares. Table 5 estimates regressions of log trade shares on railway distance. The specifications include origin-sector-year and destination-sector-year fixed effects, so identification comes from changes in bilateral railway distance after absorbing all time-varying supply conditions at the exporting province-sector level and all time-varying demand conditions at the importing province-sector level. The coefficients are negative throughout. A reduction in railway distance raises the bilateral expenditure share, with the strongest response in manufacturing. This pattern is precisely what the model predicts: lower delivered prices increase expenditure shares, and the response is stronger in sectors whose products are more tradable and more input-intensive.

Table 5. Railway Connectivity and Bilateral Trade Shares

	Dependent variable: $\ln S_{ijt}^h$				
	Pooled (1)	Agriculture (2)	Manufacturing (3)	Services (4)	Pooled (5)
$\ln \text{RailDist}_{ijt}$	-0.184*** (0.035)	-0.116** (0.047)	-0.231*** (0.052)	-0.159*** (0.043)	-0.171*** (0.037)
Highway connectivity					0.029** (0.014)
Port access					0.018* (0.010)
Province-pair fixed effects	Yes	Yes	Yes	Yes	Yes
Origin-sector-year fixed effects	Yes	Yes	Yes	Yes	Yes
Destination-sector-year fixed effects	Yes	Yes	Yes	Yes	Yes
Time-varying controls	No	No	No	No	Yes
Observations	5,220	1,740	1,740	1,740	5,220
R^2	0.971	0.958	0.976	0.965	0.973

Notes: The dependent variable is the log bilateral expenditure share. Origin-sector-year and destination-sector-year fixed effects absorb all exporter productivity, cost, and importer demand shifters at the sector-year level. Standard errors clustered by province pair are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

We next move from bilateral trade flows to regional outcomes. The model predicts that improved railway access affects real income through two channels. On the consumption side, better access to outside varieties lowers the local price index. On the production side, better market access changes local labor demand and wages. Table 6 estimates province-level regressions of wages, prices, real wages, and employment shares on the regional

railway exposure measure defined in equation (2). Columns (1)–(3) show positive effects on nominal wages, especially in manufacturing. Columns (4)–(5) show negative effects on the local price index. Columns (6)–(7) combine these effects and show that real wages rise in regions with larger railway exposure. These patterns validate the price and wage margins that drive the initial welfare gains in the structural counterfactual.

Table 6. Regional Railway Exposure, Wages, Prices, and Employment

	$\ln w_{it}^h$ Pooled (1)	$\ln w_{it}^h$ Mfg. (2)	$\ln w_{it}^h$ Services (3)	$\ln p_{it}$ Baseline (4)	$\ln p_{it}$ Controls (5)	$\ln(w_{it}^h/p_{it})$ Pooled (6)	Emp share Province (7)
<i>RailExposure_{it}</i>	0.021** (0.008)	0.028*** (0.009)	0.019** (0.008)	-0.014*** (0.004)	-0.012*** (0.004)	0.034*** (0.010)	0.006** (0.003)
GDP per capita					0.006 (0.006)		0.002 (0.002)
Capital per worker					-0.004 (0.005)		0.001 (0.001)
Province fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector fixed effects	Yes	No	No	No	No	Yes	No
Observations	1,440	480	480	480	480	1,440	480
R^2	0.743	0.761	0.728	0.816	0.824	0.779	0.692

Notes: The unit of observation is a province-sector-year in columns (1)–(3) and (6), and a province-year in columns (4), (5), and (7). *RailExposure_{it}* is constructed from bilateral railway distance changes using initial trade-share weights. Standard errors clustered by province are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Finally, we examine the migration mechanism directly. The dynamic migration block implies that workers are more likely to move toward destinations with higher future welfare and less likely to move along routes with higher migration costs. Table 7 estimates gravity-style migration regressions using the annual migration matrices constructed from census data. The dependent variable is the log odds of moving from origin i to destination j relative to staying in origin i . The main coefficient on real-wage differentials is positive, while the coefficient on recovered migration costs is negative. Columns (3)–(5) add origin-year and destination-year fixed effects, province-pair fixed effects, and controls for distance and administrative borders. The signs and magnitudes are stable. This evidence supports the use of the recovered migration costs and validates the forward-looking labor reallocation channel.

Table 7. Migration Responses to Real Income and Migration Costs

	Dependent variable: $\ln(D_{ijt}/D_{iit})$				
	Baseline (1)	Add κ (2)	Pair FE (3)	Origin/Dest. FE (4)	Full controls (5)
Real-wage differential	0.812*** (0.118)	0.774*** (0.116)	0.691*** (0.132)	0.642*** (0.129)	0.618*** (0.134)
$\ln \kappa_{ijt}$		-0.327** (0.131)	-0.301** (0.127)	-0.286** (0.121)	-0.274** (0.119)
Geographic distance					-0.043 (0.031)
Same macro region					0.056* (0.030)
Province-pair fixed effects	No	No	Yes	Yes	Yes
Origin-year fixed effects	No	No	No	Yes	Yes
Destination-year fixed effects	No	No	No	Yes	Yes
Observations	2,610	2,610	2,610	2,610	2,610
R^2	0.412	0.438	0.601	0.679	0.684

Notes: The dependent variable is the log migration odds from province i to province j relative to staying in province i . Real-wage differential is $\ln(w_{jt}/p_{jt}) - \ln(w_{it}/p_{it})$, using employment-weighted wages across sectors. Standard errors clustered by origin-destination pair are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.2 Model Fit

Having established the reduced-form patterns, we next examine whether the calibrated model reproduces the main time-series and cross-sectional features of the Chinese economy over the data period. This validation exercise is important because the counterfactual uses the same structural equations to evaluate the economy outside the observed railway path. A model that fails to reproduce the observed evolution of GDP, sectoral structure, labor allocation, and capital accumulation would not provide a credible basis for welfare analysis.

Figure 1 compares model-implied national real GDP with the observed series from 2002 to 2017. The model captures both the high-growth years before the global financial crisis and the gradual slowdown after 2010. The fit is not mechanical: while the TFP sequence is calibrated from provincial sectoral GDP growth, aggregate GDP in the model also reflects endogenous changes in trade shares, prices, migration, and capital returns. The close match therefore indicates that the dynamic general equilibrium system is internally consistent with the observed macroeconomic path.



Figure 1. National Real GDP: Model and Data

Notes: The solid line is model-implied national real GDP; circles are observed real GDP from provincial accounts aggregated to the national level. GDP is normalized to one in 2002. The model is initialized in 2002 and simulated forward using calibrated changes in productivity, trade costs, migration costs, and amenities.

Figure 2 reports the model’s performance for sectoral GDP shares. The model reproduces the decline in agriculture and the rise in services, while manufacturing remains broadly stable before declining modestly toward the end of the sample. The fit reflects the combination of sector-specific productivity trends and input-output linkages. Since manufacturing is both highly tradable and input-intensive, the model predicts that railway-induced trade-cost reductions initially support manufacturing activity, while long-run reallocation toward services emerges as income rises and the service sector becomes increasingly central in final demand.

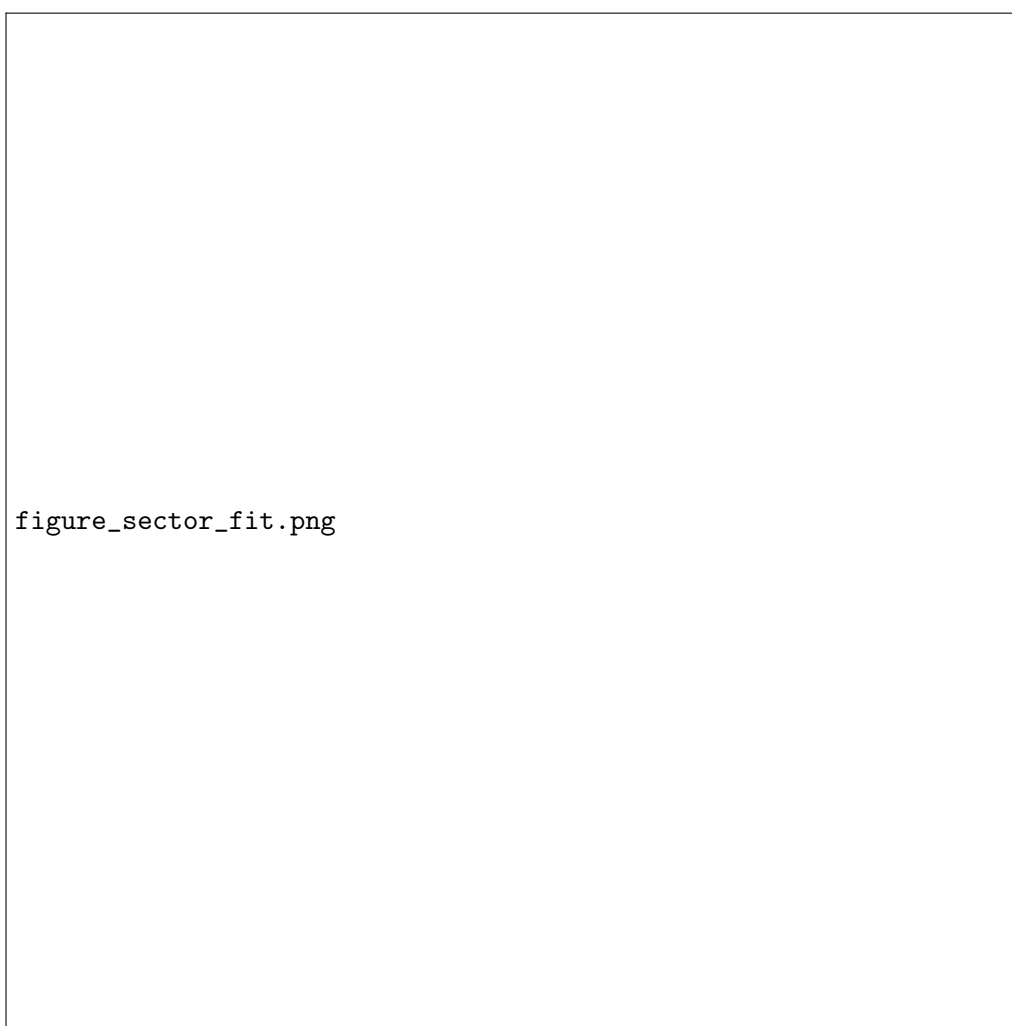


Figure 2. Sectoral GDP Shares: Model and Data

Notes: The figure compares model-implied sectoral GDP shares with observed shares for agriculture, manufacturing, and services. Solid lines denote model simulations; markers denote data. Sectoral shares are computed using real GDP by province and sector.

5.3 Aggregate Welfare Effects of Railway Construction

We now turn to the central counterfactual. The baseline economy follows the observed path of all fundamentals, including railway-induced reductions in trade costs. The counterfactual holds railway distances fixed at their 2002 levels and removes the railway-induced component of trade-cost reductions using equation (23). Productivity, amenities, and migration costs are held at their baseline paths in both economies. The welfare effect therefore isolates the contribution of railway construction operating through goods-market integration.

Table 8 reports the aggregate welfare gains. The short-run gain, measured five years after the initial year, is 1.7 percent in consumption-equivalent terms. The medium-run gain rises to 4.2 percent, and the long-run gain reaches 9.4 percent. These numbers imply substantial dynamic amplification: the long-run effect is 5.5 times the short-run effect. The interpretation is straightforward. The initial impact of railway construction lowers delivered prices and changes wages through market access. Over time, workers migrate toward regions whose real income improves, and capital accumulates in regions with higher returns. These slow-moving margins magnify the welfare consequences of the original trade-cost shock.

Table 8. Aggregate Welfare Gains from Railway Construction

Horizon	Year	Aggregate CEV	SE	Share of long-run gain
Short run	2007	1.71	0.22	18.2
Medium run	2012	4.23	0.38	45.0
End of data period	2017	6.87	0.51	73.1
Long run	Steady state	9.40	0.74	100.0
Dynamic amplification		5.50		

Notes: Welfare gains are consumption-equivalent variations in percent, computed from equation (20) and aggregated using employment weights. The counterfactual removes railway-induced trade-cost changes while holding all other fundamentals at baseline paths. Standard errors are bootstrap standard errors from resampling province pairs and recalibrating the railway elasticity.

5.4 Regional Heterogeneity

The aggregate welfare gain masks substantial regional heterogeneity. The largest gains accrue to inland provinces that experienced large improvements in railway access to coastal and central markets. The smallest gains accrue to provinces that were already well connected at the start of the sample. This pattern is consistent with the policy objective of using railway construction to integrate the interior with national markets.

Table 9 reports welfare gains by macro-region. The western and central regions experience the largest long-run gains, 12.3 and 10.7 percent respectively. The eastern region gains 6.4 percent, reflecting both its already high initial connectivity and increased competition from newly connected inland producers. The northeast gains 7.1 percent, smaller than the central and western regions but larger than the eastern average. The table also decomposes regional gains into real-wage, price-index, and option-value components. The price component is especially important in inland regions, where railway expansion lowers the cost of imported intermediate and final goods. The option-value component is also larger in these regions because improved trade integration changes future migration opportunities.

Table 9. Regional Distribution of Welfare Gains

Region	SR CEV	LR CEV	Wage comp.	Price comp.	Option-value comp.
East	1.18	6.42	2.31	2.64	1.47
Central	1.96	10.73	3.18	4.52	3.03
West	2.22	12.31	3.42	5.18	3.71
Northeast	1.41	7.08	2.54	2.81	1.73
National	1.71	9.40	2.86	3.91	2.63

Notes: Values are percent consumption-equivalent gains. Regional aggregates use employment weights. The wage and price components are computed from changes in nominal wages and consumer price indices; the option-value component captures the residual dynamic welfare component associated with future migration opportunities.

To quantify the determinants of regional heterogeneity, Table 10 regresses province-level long-run welfare gains on initial railway exposure, initial market access, sectoral composition, and income. Provinces with larger railway exposure gain more. Provinces with higher initial market access gain less, indicating diminishing returns to connectivity.

The interaction between exposure and low initial market access is positive, implying that railway construction is especially valuable for initially isolated regions. Manufacturing employment share also predicts larger gains, consistent with the importance of input-output propagation.

Table 10. Determinants of Provincial Welfare Gains

	Dependent variable: long-run provincial CEV			
	(1)	(2)	(3)	(4)
Railway exposure, 2002–2016	4.812*** (1.214)	4.206*** (1.106)	3.784*** (1.018)	3.516*** (0.982)
Initial market access		-1.327** (0.534)	-1.182** (0.511)	-1.046* (0.548)
Exposure $\times$ low market access			1.964** (0.781)	1.721** (0.746)
Manufacturing employment share, 2002				0.083** (0.036)
Log GDP per worker, 2002				-0.412 (0.368)
Macro-region fixed effects	No	No	Yes	Yes
Observations	30	30	30	30
R^2	0.418	0.563	0.681	0.724

Notes: The dependent variable is the long-run province-level consumption-equivalent welfare gain. Railway exposure is defined using initial trade-share weights. Initial market access is computed from 2002 trade shares and destination expenditures. Standard errors are heteroskedasticity robust. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.5 Sectoral Heterogeneity

The welfare gains also differ across sectors. Table 11 reports sector-level gains. Manufacturing experiences the largest long-run gain, 11.2 percent, followed by services at 8.7 percent and agriculture at 6.3 percent. The ranking is intuitive. Manufacturing is the most input-intensive sector in the calibration, with the lowest value-added share and the largest role in interregional trade. Railway-induced reductions in trade costs therefore affect manufacturing both directly, by lowering delivered costs, and indirectly, by reducing the cost of intermediate inputs. Services benefit through lower input costs and higher demand from expanding regions, while agriculture is less responsive because of its higher value-added share and lower trade intensity.

The sectoral differences also shape the regional distribution of gains. Provinces with

Table 11. Sectoral Welfare Gains and Network Exposure

Sector	SR CEV	LR CEV	Trade	Intermediate	Network	Dynamic
Agriculture	1.09	6.31	0.184	0.350	0.72	3.11
Manufacturing	2.18	11.24	0.416	0.660	3.08	5.42
Services	1.63	8.72	0.297	0.400	1.84	4.21
All sectors	1.71	9.40	0.329	0.506	2.21	5.18

Notes: Trade openness is one minus the average home expenditure share in 2002. Intermediate intensity is $1 - \gamma^h$. Network contribution is the difference between the full-model CEV and the no-input-output CEV. Dynamic contribution is the difference between long-run and short-run CEV.

larger initial manufacturing shares experience stronger wage responses because railway improvements expand the effective market for their tradable output. Provinces with larger service shares benefit more through lower local price indices and downstream demand effects. Agricultural provinces benefit primarily through cheaper intermediate and final goods rather than through large wage increases. These patterns reinforce the importance of modeling multiple sectors and input-output linkages rather than using a single-sector spatial model.

5.6 Mechanism Decomposition

We next decompose the long-run welfare gain into the mechanisms emphasized throughout the paper. The decomposition is based on restricted counterfactuals. In the no-migration counterfactual, labor allocations are fixed at their initial distribution. In the fixed-capital counterfactual, capital stocks follow their baseline path and do not respond endogenously to changes in returns. In the no-input-output counterfactual, intermediate-input linkages are removed by setting $\gamma^{h,o} = 0$ and $\gamma^h = 1$. Finally, the static counterfactual evaluates the economy before labor and capital have time to reallocate.

Table 12 reports the results. The static price and wage channels account for 1.71 percentage points, or 18.2 percent of the long-run gain. The production-network channel contributes 2.21 percentage points. This confirms that input-output linkages substantially amplify the direct effect of lower trade costs. The migration channel contributes 3.04 percentage points, reflecting the ability of workers to move toward regions whose real income and option value improve after railway expansion. The capital channel contributes 2.06

percentage points, as higher returns stimulate investment in better-connected regions. The interaction residual is small, indicating that the decomposition is approximately additive.

Table 12. Decomposition of Long-Run Welfare Gains

Channel	Contribution	Share of total	Restricted counterfactual
Static price and wage effects	1.71	18.2	Before factor adjustment
Production-network propagation	2.21	23.5	Remove input-output linkages
Labor migration	3.04	32.3	Fix labor allocation
Capital accumulation	2.06	21.9	Fix capital path
Interaction residual	0.38	4.0	Nonlinear complementarities
Total long-run gain	9.40	100.0	Full model

Notes: Contributions are measured in percentage points of consumption-equivalent welfare. The total long-run gain is the aggregate CEV in the full model. Restricted counterfactuals shut down one channel at a time.

5.7 Dynamic Adjustment

The model predicts a clear ordering of adjustment speeds. Prices respond immediately because delivered prices depend directly on trade costs. Wages respond next as expenditure shares and goods-market clearing adjust. Migration responds gradually because moving is costly and workers value future opportunities. Capital adjusts most slowly because investment responds through the Euler equation and accumulates over time.

Table 13 reports the share of long-run gains realized at different horizons. Only 18.2 percent of the long-run welfare gain is realized by 2007. By 2012, 45.0 percent is realized, and by 2017, 73.1 percent is realized. Full convergence requires roughly three decades. The price channel is the fastest: more than half of its long-run effect is realized within five years. The capital channel is the slowest: only 12.4 percent is realized within five years. The migration channel lies between these extremes.

5.8 Prices, Wages, and Market Access

The welfare effects reported above can be traced to changes in prices and wages. Railway construction lowers the delivered cost of goods, which reduces consumer price indices. At the same time, it changes the geography of demand and supply, shifting wages across sectors and regions. The net effect on real wages is generally positive but heterogeneous.

Table 13. Speed of Dynamic Adjustment

	5 years	10 years	15 years	30 years
Aggregate welfare gain	18.2	45.0	73.1	96.4
Price channel	52.7	76.4	91.2	99.1
Wage channel	31.6	58.3	81.9	97.0
Migration channel	17.9	44.7	71.8	94.2
Capital channel	12.4	33.8	69.5	92.1

Notes: Each entry reports the percent of the corresponding long-run effect realized by the horizon indicated in the column.

Table 14 reports baseline-minus-counterfactual changes in the aggregate price index, nominal wages, and real wages. The national price index is 3.7 percent lower in the long run because railway construction reduces the cost of importing goods from other provinces. Nominal wages rise by 1.6 percent on average, reflecting expanded market access and higher labor demand in connected regions. Combining these two effects, real wages rise by 5.5 percent. The real-wage gain is largest in the western region, where the price decline is large and nominal wages increase substantially from improved access to national markets.

Table 14. Effects on Prices and Wages

Region	Price index	Nominal wage	Real wage	Market access
East	-2.74	0.86	3.70	7.18
Central	-4.13	1.82	6.20	12.64
West	-4.89	2.16	7.41	15.27
Northeast	-3.02	1.13	4.28	8.46
National	-3.71	1.62	5.54	11.08

Notes: Entries are long-run baseline-minus-counterfactual percentage changes. Real wage is nominal wage divided by the local consumer price index. Market access is the model-implied expenditure-weighted access to destination markets, normalized so that entries are percentage changes relative to the counterfactual.

The price decline is not uniform across sectors. Table 15 decomposes the long-run price-index effect by sector. Manufacturing prices fall the most because manufacturing goods are the most tradable and rely heavily on traded intermediate inputs. Services prices also fall, partly because service production uses manufacturing intermediates. Agricultural prices fall the least. These sectoral patterns mirror the input-output structure in Table 2: sectors with larger intermediate-input shares experience stronger cost reductions when

railway construction lowers trade costs.

Table 15. Sectoral Price Effects

Sector	Delivered price	Sectoral p.i.	Direct trade-cost	Input-cost	Total
Agriculture	-2.18	-1.64	-1.21	-0.43	0.16
Manufacturing	-5.91	-4.72	-3.16	-1.56	2.10
Services	-3.77	-3.05	-1.72	-1.33	1.45
Aggregate		-3.71	-2.18	-1.53	3.71

Notes: Entries are long-run baseline-minus-counterfactual percentage changes. Delivered price is the expenditure-weighted average delivered price across origins and destinations. The direct trade-cost effect holds input prices fixed; the input-cost effect captures propagation through sectoral input-output linkages.

5.9 Labor Reallocation and Capital Accumulation

The model’s dynamic channels operate through labor and capital. Railway construction changes the spatial distribution of real income, which affects migration incentives, and it changes the spatial distribution of capital returns, which affects investment. These responses are slow but quantitatively important.

Table 16 reports changes in regional employment shares. In the long run, the eastern region gains 1.21 percentage points of national employment relative to the no-railway counterfactual, while the central and western regions lose 0.42 and 0.63 percentage points respectively. At first glance, this may appear inconsistent with the larger welfare gains in inland regions. The apparent tension is resolved by the model’s equilibrium logic: railway construction raises welfare in inland regions by reducing prices and improving market access, but it also raises the productivity of the national spatial system and allows some workers to relocate toward high-wage coastal markets. Welfare gains are therefore not equivalent to population gains. Infrastructure can raise welfare in a region even if some workers choose to migrate away because the option value of migration is itself part of welfare.

Capital responds differently. Table 17 shows that capital accumulates disproportionately in central and western provinces, where railway construction raises the marginal product of capital by improving access to outside markets. The western region’s long-run

Table 16. Labor Reallocation Induced by Railway Construction

Region	2007	2012	2017	Long run
East	0.24	0.63	0.94	1.21
Central	-0.09	-0.24	-0.33	-0.42
West	-0.12	-0.31	-0.49	-0.63
Northeast	-0.03	-0.08	-0.12	-0.16

Notes: Entries are baseline-minus-counterfactual changes in regional employment shares, measured in percentage points. Employment is summed across sectors.

capital stock is 6.8 percent higher than in the no-railway counterfactual, compared with 3.4 percent in the eastern region. This capital response partly offsets labor outflows from some inland regions and helps sustain the large welfare gains there.

Table 17. Capital Accumulation Induced by Railway Construction

Region	Capital stock	Investment return	Capital-output ratio	Output
East	3.42	0.18	0.06	4.21
Central	5.91	0.31	0.11	6.48
West	6.84	0.36	0.13	7.24
Northeast	3.76	0.22	0.07	4.63
National	4.92	0.27	0.09	5.61

Notes: Entries are long-run baseline-minus-counterfactual percentage changes, except investment return and capital-output ratio, which are reported in level changes.

5.10 Production-Network Propagation

The production network is a key amplifier of railway gains. A reduction in the delivered price of manufactured inputs lowers the marginal cost of sectors that use those inputs. These sectors then reduce their own output prices, affecting downstream users in subsequent rounds. Because the model includes three broad sectors rather than a full input-output matrix, the network is parsimonious, but it captures the main pattern visible in China's input-output table: manufacturing is both a major user and supplier of intermediate inputs.

To quantify propagation, Table 18 reports the contribution of each source-sector price decline to each user-sector cost reduction. Manufacturing-to-manufacturing propagation is the largest cell, reflecting the high within-manufacturing intermediate share. Manufacturing-

to-services is also important, indicating that services benefit materially from cheaper manufactured inputs. Agriculture has smaller network spillovers, both because it uses fewer intermediates and because it accounts for a smaller share of final consumption.

Table 18. Input-Output Propagation of Railway-Induced Cost Reductions

User sector	Source of input-cost reduction			Total cost reduction
	Agriculture	Manufacturing	Services	
Agriculture	0.14	0.38	0.09	0.61
Manufacturing	0.07	1.74	0.31	2.12
Services	0.03	0.82	0.44	1.29
Total	0.24	2.94	0.84	4.02

Notes: Entries report percentage-point contributions to long-run cost reductions from input-output propagation. Rows are user sectors; columns are source sectors whose prices decline due to railway-induced trade-cost reductions.

5.11 Robustness

We assess the robustness of the results along four dimensions: structural parameters, alternative railway elasticities, alternative counterfactual definitions, and restricted model environments. Table A.2 summarizes the results. The aggregate long-run welfare gain remains between 7.8 and 11.2 percent across all specifications. The qualitative conclusions are unchanged: welfare gains are positive and sizable; inland regions gain more than coastal regions; dynamic margins account for most of the long-run effect; and production networks amplify the gains.

The parameter sensitivities have intuitive patterns. A higher trade elasticity increases the reallocation of trade shares and strengthens the production-network channel, but it also reduces the love-of-variety component of trade gains. A lower migration dispersion parameter increases the responsiveness of workers to real-income differences, raising the migration contribution. A lower depreciation rate makes capital adjustment more persistent, while a higher depreciation rate accelerates capital turnover but reduces the long-run contribution of capital accumulation. None of these parameter variations overturns the baseline conclusion that dynamic adjustment is quantitatively central.

The alternative counterfactuals clarify which parts of the railway program matter most.

Removing only the post-2008 high-speed rail expansion generates a long-run welfare gain of 5.21 percent relative to the no-HSR counterfactual, implying that roughly half of the total railway gain is attributable to the high-speed era and the associated reallocation of conventional rail capacity. The shift-share attribution produces a slightly smaller aggregate gain than the baseline, as expected, because it assigns to railways only the component of trade-cost decline most strongly correlated with observed within-pair railway improvements. The fact that the estimate remains close to the baseline suggests that the main results are not driven by over-attributing general transport improvements to railways.

The restricted model environments also reinforce the paper’s main message. Without input-output linkages, the long-run gain falls from 9.40 to 7.19 percent. Without migration, it falls to 6.36 percent. With capital fixed, it falls to 7.34 percent. These reductions are large relative to the baseline and demonstrate that each of the three model extensions—production networks, forward-looking migration, and endogenous capital accumulation—contributes materially to the welfare calculation.

5.12 Alternative Measures of Railway Exposure

The baseline counterfactual attributes trade-cost changes to railway improvements using initial trade-share weights. This is natural in the model because trade shares determine the economic relevance of each bilateral connection. Nevertheless, one concern is that the results might depend on this weighting scheme. We therefore construct three alternative exposure measures: equal weights across destination provinces, destination population weights, and destination GDP weights. Table 19 reports the results.

Table 19. Robustness to Alternative Railway Exposure Measures

Exposure measure	Aggregate	East	Central	West	Northeast
Initial trade-share weights	9.40	6.42	10.73	12.31	7.08
Equal weights	8.91	6.29	10.04	11.73	6.81
Destination population weights	9.17	6.35	10.41	12.02	6.94
Destination GDP weights	9.02	6.51	10.18	11.64	6.88

Notes: Entries report long-run consumption-equivalent welfare gains in percent. Each row recomputes the railway-induced component of trade costs using a different measure of bilateral exposure.

The aggregate welfare gain remains between 8.9 and 9.6 percent across the four measures. Regional patterns are also stable: central and western provinces gain more than eastern provinces under all exposure definitions. The trade-share-weighted measure produces the largest estimate, which is expected because it emphasizes economically relevant corridors. Equal weights produce the smallest estimate because many bilateral railway improvements connect province pairs with limited trade volumes.

5.13 Summary of Findings

The empirical and quantitative results support a coherent interpretation of China’s railway expansion. Reduced-form estimates show that railway improvements are associated with lower recovered trade costs, larger bilateral trade shares, lower regional prices, higher real wages, and migration responses consistent with real-income differentials. The calibrated model reproduces the observed evolution of national GDP, sectoral transformation, employment redistribution, and capital accumulation during 2002–2017. In the structural counterfactual, railway construction generates a long-run welfare gain of 9.4 percent, with substantially larger gains in central and western provinces than in initially well-connected eastern provinces.

The mechanism decomposition reveals why the gains are large. The static price and wage effect is positive but accounts for less than one-fifth of the long-run welfare gain. Production-network propagation amplifies the initial cost reductions. Forward-looking migration allows workers to benefit from improved future opportunities, and capital accumulation raises productive capacity in regions whose returns increase after railway integration. These dynamic margins operate slowly, implying that short-horizon evaluations substantially understate the value of infrastructure. The results therefore speak directly to the broader question motivating the paper: transport infrastructure matters not only because it lowers the cost of moving goods, but because it reshapes the dynamic allocation of labor and capital across space.

6 Conclusion

This paper quantifies the welfare effects of China’s 2002–2016 railway expansion using a dynamic spatial general equilibrium model with multi-sector production networks, forward-looking migration, and regional capital accumulation. Calibrating the model to provincial data and exploiting within-pair variation in railway shortest-path distances, we find that railway construction generated an aggregate long-run welfare gain of 9.4 percent in consumption-equivalent terms, with central and western provinces gaining 10.7 and 12.3 percent respectively, roughly twice as much as the already well-connected eastern region, which gains 6.4 percent. Only 1.7 percent of this long-run effect is realized within five years and 6.9 percent by the end of the data window in 2017, implying a dynamic amplification factor of 5.5 between the short and long-run gains. A mechanism decomposition reveals that the static price and wage channels account for only 18.2 percent of the total gain, while production-network propagation (23.5 percent), labor reallocation (32.3 percent), and capital accumulation (21.9 percent) together contribute the remaining four-fifths. These findings imply that conventional short-horizon evaluations, whether reduced-form estimates or static general equilibrium calculations, understate the true returns to large-scale transport infrastructure by a factor of more than five, and that the regional convergence effects of railway construction operate primarily through slow-moving factor reallocation rather than through immediate price adjustments.

Three methodological contributions underlie these results. First, extending the dynamic spatial framework of [Kleinman et al. \(2023\)](#) to a multi-sector input–output environment reveals that network propagation contributes 2.21 percentage points to the long-run welfare gain, a margin that varies systematically across sectors and is largest in manufacturing (11.2 percent long-run CEV) because of its deep backward linkages and high tradability. Second, combining structural trade-cost recovery with a reduced-form railway elasticity estimated from within-pair variation provides a transparent and internally consistent bridge between the quasi-experimental variation in the data and the general equilibrium counterfactual, addressing the endogenous placement of railway routes without sacrificing the richness of the structural model. Third, the sequential calibration

procedure, which recovers trade costs, migration costs, productivity, and amenities in a logically ordered chain without requiring the levels of unobservable fundamentals, offers a replicable template for quantifying dynamic spatial models in settings where rich panel data exist alongside substantial unobserved heterogeneity.

Several important questions remain open. Our framework treats railway construction as exogenous, but a complete political-economy model of infrastructure allocation, in which investment responds to regional lobbying, fiscal capacity, and strategic planning, could substantially alter the counterfactual distribution of gains. Allowing for endogenous firm entry and selection, as in [Melitz \(2003\)](#), would capture the productivity composition effects of improved market access that our representative-firm structure abstracts from. Finally, incorporating congestion externalities and agglomeration economies into the amenity block would enrich the model's predictions for the long-run spatial distribution of population, a dimension along which the current framework, by treating local amenities as exogenous, imposes strong parametric restrictions. We view these extensions as natural next steps for a research program aimed at providing rigorous welfare evaluations of the infrastructure investments that continue to reshape economic geography across the developing world.

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A Derivations of Key Model Equations

This appendix provides detailed derivations of the main equations used in the model. We cover cost minimization and the factory-gate price, the CES price index and trade shares, the migration probability and welfare recursion, the Euler equation and consumption-share recursion, the goods-market clearing condition, the Gumbel-logit closed form, and the relationship between value-added and gross-output TFP.

A.1 Cost Minimization and Factory-Gate Price

The firm in region i sector h minimizes total cost $w_{it}^h l_{it}^h + r_{it} k_{it}^h + \sum_o p_{it}^o x_{it}^{h,o}$ subject to the production function (3). The Lagrangian is:

$$\mathcal{L} = w_{it}^h l_{it}^h + r_{it} k_{it}^h + \sum_o p_{it}^o x_{it}^{h,o} - \lambda \left[z_{it}^h \left(\frac{l_{it}^h}{\mu^h \gamma^h} \right)^{\mu^h \gamma^h} \left(\frac{k_{it}^h}{(1-\mu^h) \gamma^h} \right)^{(1-\mu^h) \gamma^h} \prod_o \left(\frac{x_{it}^{h,o}}{\gamma^{h,o}} \right)^{\gamma^{h,o}} - y_{it}^h \right]. \quad (\text{A.1})$$

The first-order conditions give $w_{it}^h l_{it}^h = \lambda \mu^h \gamma^h y_{it}^h$, $r_{it} k_{it}^h = \lambda (1-\mu^h) \gamma^h y_{it}^h$, and $p_{it}^o x_{it}^{h,o} = \lambda \gamma^{h,o} y_{it}^h$. Under perfect competition, $\lambda = p_{it}^h$ (price equals marginal cost). The exponents sum to one: $\mu^h \gamma^h + (1-\mu^h) \gamma^h + \sum_o \gamma^{h,o} = \gamma^h + (1-\gamma^h) = 1$, so factor payments exhaust revenue (zero profits). Substituting the optimal inputs back into (3), y_{it}^h cancels on both sides, yielding:

$$p_{it}^h = \frac{1}{z_{it}^h} (w_{it}^h)^{\mu^h \gamma^h} r_{it}^{(1-\mu^h) \gamma^h} \prod_o (p_{it}^o)^{\gamma^{h,o}} = \frac{C_{it}^h}{z_{it}^h}. \quad (\text{A.2})$$

The delivered price is then $p_{ijt}^h = \tau_{ijt}^h C_{jt}^h / z_{jt}^h$, since shipping from source j inherits j 's cost structure and incurs iceberg cost τ_{ijt}^h .

A.2 CES Price Index and Trade Shares

The CES expenditure-minimization problem is $\min \sum_j p_{ijt}^h c_{ijt}^h$ subject to $[\sum_j (c_{ijt}^h)^{\theta/(\theta+1)}]^{(\theta+1)/\theta} = c_{it}^h$. The first-order conditions give, for any two sources j, m , $c_{ijt}^h / c_{imt}^h = (p_{ijt}^h / p_{imt}^h)^{-(1+\theta)}$. Substituting the demand function $c_{ijt}^h = c_{it}^h (p_{ijt}^h / p_{it}^h)^{-(1+\theta)}$ into the expenditure identity $E_{it}^h = \sum_j p_{ijt}^h c_{ijt}^h$ and requiring $E_{it}^h = p_{it}^h c_{it}^h$ yields:

$$p_{it}^h = \left[\sum_{j=1}^N (p_{ijt}^h)^{-\theta} \right]^{-1/\theta}, \quad S_{ijt}^h = \frac{p_{ijt}^h c_{ijt}^h}{E_{it}^h} = \left(\frac{p_{ijt}^h}{p_{it}^h} \right)^{-\theta}, \quad (\text{A.3})$$

with $\sum_j S_{ijt}^h = 1$ by construction.

A.3 Migration Probability and Welfare Recursion

The worker in (i, h) at t chooses destination j to maximize $\beta E_t[V_{j,t+1}^h] - \ln \kappa_{ijt} + \rho \varepsilon_{jt}$, with $\varepsilon_{jt} \stackrel{\text{i.i.d.}}{\sim} \text{Gumbel}(0, 1)$. By the standard Gumbel result (see Section A.6), the choice probability and expected maximum are:

$$D_{ijt}^h = \frac{\exp\{(\beta E_t[V_{j,t+1}^h] - \ln \kappa_{ijt})/\rho\}}{\sum_m \exp\{(\beta E_t[V_{m,t+1}^h] - \ln \kappa_{imt})/\rho\}} = \frac{u_{j,t+1}^h / \kappa_{ijt}^{1/\rho}}{\sum_m u_{m,t+1}^h / \kappa_{imt}^{1/\rho}}, \quad (\text{A.4})$$

$$E_\varepsilon \left[\max_j \{\cdot\} \right] = \rho \ln \sum_j \exp \left(\frac{\beta E_t[V_{j,t+1}^h] - \ln \kappa_{ijt}}{\rho} \right) + \rho \gamma_E, \quad (\text{A.5})$$

where $u_{it}^h \equiv \exp(\beta E_t[V_{it}^h]/\rho)$ and $\gamma_E \approx 0.5772$ is the Euler–Mascheroni constant (absorbed into amenities). Substituting into $V_{it}^h = \ln(b_{it} w_{it}^h / p_{it}) + E_\varepsilon[\max_j \{\cdot\}]$ and exponentiating by β/ρ yields the welfare recursion (12):

$$u_{it}^h = \left[b_{it} \frac{w_{it}^h}{p_{it}} \left(\sum_j \frac{u_{j,t+1}^h}{\kappa_{ijt}^{1/\rho}} \right)^\rho \right]^{\beta/\rho}. \quad (\text{A.6})$$

A.4 Euler Equation and Consumption-Share Recursion

The CRRA Euler equation is $(c_{it}^k)^{-1/\psi} = \beta E_t[R_{i,t+1}(c_{i,t+1}^k)^{-1/\psi}]$. Substituting $c_{it}^k = \varsigma_{it} R_{it} k_{it}$ and $k_{i,t+1} = (1 - \varsigma_{it}) R_{it} k_{it}$, then cancelling $(R_{it} k_{it})^{-1/\psi}$:

$$\varsigma_{it}^{-1/\psi} = \beta (1 - \varsigma_{it})^{-1/\psi} E_t \left[R_{i,t+1}^{1-1/\psi} \varsigma_{i,t+1}^{-1/\psi} \right]. \quad (\text{A.7})$$

Taking the ψ -th power and rearranging gives $\varsigma_{it}^{-1} = 1 + \beta^\psi \{E_t[\varsigma_{i,t+1}^{-1/\psi} R_{i,t+1}^{1-1/\psi}]\}^\psi$. Under perfect foresight this simplifies to (16). For the backward recursion used in the solver, we solve explicitly for ς_{it} :

$$\varsigma_{it} = \frac{\varsigma_{i,t+1}}{\varsigma_{i,t+1} + \beta^\psi R_{i,t+1}^{\psi-1}}. \quad (\text{A.8})$$

In the steady state ($\varsigma = \bar{\varsigma}$, $R = 1/\beta$), (A.8) implies $\bar{\varsigma} = 1 - \beta$, which we use as a terminal condition.

A.5 Goods-Market Clearing

From (4), gross output revenue is $p_{j,t}^o y_{j,t}^o = W_{j,t}^o / (\mu^o \gamma^o)$ and value added is $W_{j,t}^o / \mu^o$. In balanced trade, final-consumption expenditure equals total value added: $E_{j,t}^{\text{final}} = \sum_o W_{j,t}^o / \mu^o$. Sector o in region j spends $\gamma^{o,h} W_{j,t}^o / (\mu^o \gamma^o)$ on sector- h intermediates. Combining final and intermediate demand:

$$E_{j,t}^h = \psi^h \sum_o \frac{W_{j,t}^o}{\mu^o} + \sum_o \gamma^{o,h} \frac{W_{j,t}^o}{\mu^o \gamma^o} = \sum_{o=1}^J \frac{\psi^h \gamma^o + \gamma^{o,h}}{\mu^o \gamma^o} W_{j,t}^o. \quad (\text{A.9})$$

Market clearing for source (i, h) requires $W_{i,t}^h / (\mu^h \gamma^h) = \sum_j S_{j,i,t}^h E_{j,t}^h$.

A.6 Gumbel Distribution and Logit Choice Probabilities

Consider N options with payoffs $V_j = \bar{V}_j + \rho \varepsilon_j$, where $\varepsilon_j \stackrel{\text{i.i.d.}}{\sim} \text{Gumbel}(0, 1)$ with CDF $F(\varepsilon) = \exp(-e^{-\varepsilon})$ and density $f(\varepsilon) = e^{-\varepsilon} \exp(-e^{-\varepsilon})$. The probability that option j is chosen is:

$$\Pr[j = \arg \max_m V_m] = \int_{-\infty}^{\infty} \prod_{m \neq j} F\left(\varepsilon_j + \frac{\bar{V}_j - \bar{V}_m}{\rho}\right) f(\varepsilon_j) d\varepsilon_j = \frac{e^{\bar{V}_j / \rho}}{\sum_{m=1}^N e^{\bar{V}_m / \rho}}. \quad (\text{A.10})$$

The expected maximum (log-sum formula) is:

$$E\left[\max_m V_m\right] = \rho \ln \sum_{m=1}^N e^{\bar{V}_m / \rho} + \rho \gamma_E. \quad (\text{A.11})$$

Applying this to the migration problem with $\bar{V}_j = \beta E_t[V_{j,t+1}^h] - \ln \kappa_{ijt}$ yields the migration probability (13) in the main text.

A.7 Value-Added TFP and Gross-Output TFP

The TFP $z_{i,t}^h$ in the model (3) is a gross-output residual, whereas sectoral Solow residuals constructed from national accounts are typically value-added based. Let VA denote value added and M total intermediate input. Under Cobb–Douglas with value-added share γ^h and value-added TFP $z^{VA,h}$:

$$Y = z \cdot VA^\gamma M^{1-\gamma}, \quad VA = z^{VA} L^\mu K^{1-\mu}. \quad (\text{A.12})$$

Log-differentiating and using the cost-minimizing condition $p_M M = (1 - \gamma)pY$:

$$\Delta \ln Y = \Delta \ln z + \gamma \Delta \ln VA + (1 - \gamma) \Delta \ln M, \quad \Delta \ln VA = \Delta \ln z^{VA} + \mu \Delta \ln L + (1 - \mu) \Delta \ln K. \quad (\text{A.13})$$

Following [Basu and Fernald \(1997\)](#), once intermediate inputs adjust optimally, the relationship between gross-output and value-added TFP changes reduces to

$$\Delta \ln z_{it}^h = \gamma^h \cdot \Delta \ln z_{it}^{VA,h}. \quad (\text{A.14})$$

Intuitively, value added only constitutes the fraction γ^h of gross output, so a 1% increase in value-added TFP translates into a $\gamma^h\%$ increase in gross-output TFP once intermediate-input adjustment is accounted for. We use [\(A.14\)](#) to convert national sectoral Solow residuals into the common trend A_t^h that anchors the calibration.

B Identification of Unobserved Fundamentals

This appendix collects the closed-form inversions used to recover trade costs, migration costs, relative welfare, labor income, and relative productivity from observed data.

B.1 Head–Ries Ratio for Trade Costs

From the trade share $S_{ijt}^h = (\tau_{ijt}^h C_{jt}^h / z_{jt}^h)^{-\theta} / \sum_m (\tau_{imt}^h C_{mt}^h / z_{mt}^h)^{-\theta}$ and $\tau_{iit}^h = 1$:

$$\frac{S_{ijt}^h}{S_{iit}^h} = (\tau_{ijt}^h)^{-\theta} \left(\frac{C_{jt}^h / z_{jt}^h}{C_{it}^h / z_{it}^h} \right)^{-\theta}, \quad \frac{S_{jtt}^h}{S_{jtt}^h} = (\tau_{jtt}^h)^{-\theta} \left(\frac{C_{it}^h / z_{it}^h}{C_{jt}^h / z_{jt}^h} \right)^{-\theta}. \quad (\text{A.15})$$

Multiplying the two and using $\tau_{ijt}^h = \tau_{jtt}^h$: all cost and productivity terms cancel, yielding

$$\tau_{ijt}^h = \left(\frac{S_{ijt}^h S_{jtt}^h}{S_{iit}^h S_{jtt}^h} \right)^{-1/(2\theta)}. \quad (\text{A.16})$$

B.2 Analogous Ratio for Migration Costs

From $D_{ijt}^h / D_{iit}^h = (u_{j,t+1}^h / u_{i,t+1}^h) \kappa_{ijt}^{-1/\rho}$ and its symmetric counterpart, multiplying eliminates the welfare terms:

$$\kappa_{ijt} = \left(\frac{D_{ijt}^h D_{jtt}^h}{D_{iit}^h D_{jtt}^h} \right)^{-\rho/2}. \quad (\text{A.17})$$

Since κ_{ijt} does not depend on h , we recover a separate estimate from each sector and take the employment-weighted geometric mean to reduce noise.

B.3 Relative Welfare: Perron–Frobenius Eigenvector

Writing $u_{it}^h = U_t^h \hat{u}_{it}^h$ with $\sum_i \hat{u}_{it}^h = 1$ and setting $j = i$ in (13) so that $\kappa_{iit} = 1$:

$$\hat{u}_{j,t+1}^h = \sum_{m=1}^N \frac{D_{jjt}^h}{\kappa_{jmt}^{1/\rho}} \hat{u}_{m,t+1}^h. \quad (\text{A.18})$$

Define $\tilde{D}_t^h$ with $[\tilde{D}_t^h]_{jm} = D_{jjt}^h / \kappa_{jmt}^{1/\rho}$. Then $\hat{\mathbf{U}}_{t+1}^h$ is the unit-eigenvalue right eigenvector of $\tilde{D}_t^h$, which is unique (up to normalization) by Perron–Frobenius since $\tilde{D}_t^h$ has non-negative entries.

B.4 Labor Income: Fixed-Point System

Stacking (19) into an NJ -vector $\mathbf{W}_t$ gives $\mathbf{W}_t = A_t^W \mathbf{W}_t$, where

$$[A_t^W]_{(i,h),(j,o)} = \mu^h \gamma^h \cdot \frac{\psi^h \gamma^o + \gamma^{o,h}}{\mu^o \gamma^o} \cdot S_{jit}^h. \quad (\text{A.19})$$

The unique positive eigenvector is pinned down in scale by the national-labor-compensation anchor $\sum_{i,h} W_{it}^h = L_t^{\text{nom}}$. Wages are recovered as $w_{it}^h = W_{it}^h / l_{it}^h$ and the capital rental as $r_{it} = k_{it}^{-1} \sum_h [(1 - \mu^h) / \mu^h] W_{it}^h$.

B.5 Relative Productivity: Nested Eigenvector Iteration

Writing $z_{it}^h = A_t^h \hat{z}_{it}^h$, the common trend A_t^h cancels from trade shares, so $\hat{z}_{it}^h$ is identified by the home-share condition. Setting $i = j$ in the trade share and defining $\tilde{z}_{it}^h \equiv (\hat{z}_{it}^h)^\theta$:

$$\tilde{z}_{it}^h = \sum_{m=1}^N S_{iit}^h \left(\frac{\tau_{imt}^h C_{mt}^h}{C_{it}^h} \right)^{-\theta} \tilde{z}_{mt}^h. \quad (\text{A.20})$$

This is an eigenvalue-1 problem solved jointly with the price index $p_{it}^h = (C_{it}^h / z_{it}^h) (S_{iit}^h)^{1/\theta}$ via nested iteration: given a guess $\hat{z}_{it}^{h,(n)}$, update $p_{it}^{h,(n)}$; plug into C_{it}^h ; solve the eigenvalue problem for $\hat{z}_{it}^{h,(n+1)}$; repeat until $\max_{i,h} |\hat{z}_{it}^{h,(n+1)} - \hat{z}_{it}^{h,(n)}| < 10^{-10}$. Finally, normalize by the geometric mean: $\hat{z}_{it}^h \leftarrow \hat{z}_{it}^h / \exp(N^{-1} \sum_m \ln \hat{z}_{mt}^h)$.

B.6 Common Trends and Amenity Decomposition

Common productivity trend A_t^h . With $A_0^h = 1$, we anchor A_t^h to national sectoral Solow residuals:

$$\ln A_t^h - \ln A_0^h = \gamma^h \left(\ln TFP_t^{VA,h} - \ln TFP_0^{VA,h} \right), \quad (\text{A.21})$$

where $TFP_t^{VA,h} = RVA_t^h / [(L_t^h)^{\mu^h} (K_t^h)^{1-\mu^h}]$, RVA_t^h is real value added, and the γ^h factor comes from (A.14).

Price scaling. The raw price indices p_{it}^{raw} from the calibrated z_{it}^h and C_{it}^h are normalized by the national CPI anchor (9). Defining $P_t^{\text{model}} = \prod_i (p_{it})^{\varpi_{it}}$ with population weights $\varpi_{it} = L_{it} / \sum_m L_{mt}$ and $P_t^{\text{data}} = CPI_t / CPI_0$, the uniform scaling factor $\Lambda_t = P_t^{\text{data}} / P_t^{\text{raw}}$ is applied: $p_{it} = p_{it}^{\text{raw}} \Lambda_t$.

Amenity decomposition. Write $b_{it} = B_t \hat{b}_{it}$ with $N^{-1} \sum_i \ln \hat{b}_{it} = 0$. Invert the welfare equation for the raw residual:

$$\widetilde{\ln b_{it}}^h \equiv \frac{\rho}{\beta} \ln \hat{u}_{it}^h - \ln \frac{w_{it}^h}{p_{it}} - \rho \ln \sum_j \frac{\hat{u}_{j,t+1}^h}{\kappa_{ijt}^{1/\rho}}, \quad (\text{A.22})$$

and aggregate across sectors with employment weights: $\widetilde{\ln b_{it}}^h = \sum_h (l_{it}^h / L_{it}) \widetilde{\ln b_{it}}^h$. Project onto observable quality-of-life indicators X_{it} (air quality, medical and educational resources, transport density, climate, public safety):

$$\widetilde{\ln b_{it}}^h = \alpha_i + \beta_b' X_{it} + \varepsilon_{it}, \quad (\text{A.23})$$

estimated by WLS with weights L_{it} . The common trend is $\ln B_t \equiv \hat{\beta}_b' \bar{X}_t$ with $\bar{X}_t = \sum_i \varpi_{it} X_{it}$, and the relative component is $\ln \hat{b}_{it} = \hat{\alpha}_i + \hat{\beta}_b' (X_{it} - \bar{X}_t) + \hat{\varepsilon}_{it}$, demeaned cross-sectionally to enforce $\sum_i \ln \hat{b}_{it} = 0$.

Common welfare trend U_t^h . Given all other objects, U_t^h is recovered by backward recursion from $U_T^h = 1$:

$$\ln U_t^h = \beta \ln U_{t+1}^h + \sum_i \omega_{it}^h \left[\frac{\beta}{\rho} \left(\ln B_t + \ln \hat{b}_{it} + \ln \frac{w_{it}^h}{p_{it}} + \rho \ln \sum_j \frac{\hat{u}_{j,t+1}^h}{\kappa_{ijt}^{1/\rho}} \right) - \ln \hat{u}_{it}^h \right], \quad (\text{A.24})$$

with $\omega_{it}^h = l_{it}^h / \sum_m l_{mt}^h$. Because $\beta < 1$, backward iteration is numerically stable.

C Calibration Procedure

The full calibration follows a sequential 10-step procedure. Each step uses only quantities determined in earlier steps or observed directly, ensuring logical consistency.

1. **Bilateral trade costs τ_{ijt}^h :** apply the Head–Ries ratio to MRIO trade shares for benchmark years 2002, 2007, 2012, 2017; interpolate with monotone cubic Hermite

splines; apply Gaussian smoothing ($\sigma = 2$) to remove high-frequency noise; bound $\tau \in [1, 5]$.

2. **Bilateral migration costs** κ_{ijt} : apply the analogous ratio to census-based annual migration matrices (from matrix root extraction, see ??); linear interpolation in $\ln \kappa$ across census years; backward extrapolation for pre-2005 years; hold constant post-2015.
3. **Relative welfare** $\hat{u}_{it}^h$: solve the Perron–Frobenius eigenvector problem in Section B.3 with normalization $\sum_i \hat{u}_{it}^h = 1$.
4. **Labor income and wages** W_{it}^h, w_{it}^h : solve the fixed-point system in Section B.4, scale-anchored by national labor compensation L_t^{nom} ; recover $w_{it}^h = W_{it}^h/l_{it}^h$.
5. **Capital rental rate** r_{it} : compute $r_{it} = k_{it}^{-1} \sum_h [(1 - \mu^h)/\mu^h] W_{it}^h$ using PIM-based capital stocks.
6. **Relative productivity** $\hat{z}_{it}^h$: solve the nested eigenvector–price iteration in Section B.5; geometric-mean normalization.
7. **Common productivity trend and prices** A_t^h, p_{it}^h, p_{it} : anchor A_t^h to national sectoral Solow residuals via (A.14); compute raw prices and apply uniform CPI scaling Λ_t .
8. **Amenity decomposition** $B_t, \hat{b}_{it}, b_{it}$: invert the welfare equation for $\widetilde{\ln b_{it}}$; project onto observable quality-of-life indicators; decompose into common trend and relative component.
9. **Common welfare trend** U_t^h, u_{it}^h : backward recursion from $U_T^h = 1$.
10. **Change rates**: compute $\dot{\tau}_{ijt}^h, \dot{\kappa}_{ijt}, \dot{z}_{it}^h, \dot{b}_{it}$ from adjacent-period ratios; set all change rates to 1 beyond the data window.

D Exact-Hat Algebra: Complete System and Solution Algorithm

D.1 Complete System in Changes

For any variable x_{it} , let $\dot{x}_{i,t+1} \equiv x_{i,t+1}/x_{it}$. The full system of dynamic equations in changes is:

1. **Migration probability:** $\dot{D}_{ij,t+1}^h = \frac{\dot{u}_{j,t+2}^h/\dot{\kappa}_{ij,t+1}^{1/\rho}}{\sum_m D_{imt}^h \dot{u}_{m,t+2}^h/\dot{\kappa}_{im,t+1}^{1/\rho}}$
2. **Labor evolution:** $l_{j,t+1}^h = \sum_i D_{ijt}^h l_{it}^h$
3. **Trade price:** $\dot{p}_{ij,t+1}^h = (\dot{\tau}_{ij,t+1}^h/\dot{z}_{j,t+1}^h)[(\dot{w}_{j,t+1}^h)(\dot{r}_{j,t+1})^{1-\mu^h}]^{\gamma^h} \prod_o (\dot{p}_{j,t+1}^o)^{\gamma^{h,o}}$
4. **Sectoral price index:** $\dot{p}_{i,t+1}^h = [\sum_j S_{ijt}^h (\dot{p}_{ij,t+1}^h)^{-\theta}]^{-1/\theta}$
5. **Aggregate price:** $\dot{p}_{i,t+1} = \prod_h (\dot{p}_{i,t+1}^h)^{\psi^h}$
6. **Trade share update:** $S_{ji,t+1}^h/S_{jit}^h = (\dot{p}_{ji,t+1}^h/\dot{p}_{j,t+1}^h)^{-\theta}$
7. **Goods-market clearing in changes:** $\dot{W}_{i,t+1}^h = \sum_{j,o} \omega_{jit}^{h,o} (S_{ji,t+1}^h/S_{jit}^h) \dot{W}_{j,t+1}^o, \omega_{jit}^{h,o} = \frac{[(\psi^h \gamma^o + \gamma^{o,h})/(\mu^o \gamma^o)] S_{jit}^h W_{jt}^o}{W_{it}^h/(\mu^h \gamma^h)}$
8. **Wage change:** $\dot{w}_{i,t+1}^h = \dot{W}_{i,t+1}^h/\dot{l}_{i,t+1}^h$
9. **Capital return:** $\dot{r}_{i,t+1} = (\sum_h \phi_{it}^h \dot{W}_{i,t+1}^h)/\dot{k}_{i,t+1}, \phi_{it}^h = \frac{[(1-\mu^h)/\mu^h] W_{it}^h}{\sum_o [(1-\mu^o)/\mu^o] W_{it}^o}$
10. **Capital accumulation:** $k_{i,t+1} = (1 - \varsigma_{it}) R_{it} k_{it}$
11. **Investment return:** $R_{i,t+1} = 1 - \delta + (R_{it} - 1 + \delta) \dot{r}_{i,t+1}/\dot{p}_{i,t+1}$
12. **Welfare change (backward):** $\dot{u}_{i,t+1}^h = (\dot{b}_{i,t+1} \dot{w}_{i,t+1}^h/\dot{p}_{i,t+1})^{\beta/\rho} [\sum_j D_{ijt}^h \dot{u}_{j,t+2}^h/\dot{\kappa}_{ij,t+1}^{1/\rho}]^\beta$
13. **Consumption share (backward):** $\varsigma_{it} = \varsigma_{i,t+1}/(\varsigma_{i,t+1} + \beta^\psi R_{i,t+1}^{\psi-1})$

Terminal conditions: $\dot{u}_{i,T+1}^h = 1$, $R_{iT} = 1/\beta$, $\varsigma_{iT} = 1 - \beta$, where T is the terminal horizon (long enough for the economy to approach steady state; we use $T = 50$ years).

D.2 Forward–Backward Solution Algorithm

Initialization. Given calibrated initial objects $\{w_{i0}^h, r_{i0}, p_{i0}^h, p_{i0}, l_{i0}^h, k_{i0}, k_{i1}, S_{ij0}^h, D_{ij0}^h, W_{i0}^h\}$, compute $R_{i0} = 1 - \delta + r_{i0}/p_{i0}$, $\omega_{ji0}^{h,o}$, ϕ_{i0}^h . Guess initial sequences $\{\dot{u}_{it}^{h,(0)}\}_{t=1}^{T+1}$ and $\{\varsigma_{it}^{(0)}\}_{t=0}^T$ satisfying the terminal conditions.

Outer loop (iteration n). Alternate between a forward pass and a backward pass:

Forward pass. For $t = 0, 1, \dots, T - 1$:

1. Update migration probabilities $D_{ij,t+1}^h$ using equation (1) with current $\{u^{h,(n)}\}$.
2. Update labor allocation $l_{i,t+1}^h$ via equation (2).
3. *Inner fixed point for static equilibrium.* Solve equations (3)–(9) jointly:
 - (a) Initialize $\dot{p}_{i,t+1}^{h,(0)} = 1$.
 - (b) Compute $\dot{p}_{ij,t+1}^h$ from (3) using current $\{\dot{w}_{j,t+1}^h, \dot{r}_{j,t+1}, \dot{p}_{j,t+1}^o\}$.
 - (c) Update $\dot{p}_{i,t+1}^h$ via (4) and trade-share changes via (6).
 - (d) Solve the linear system (7) for $\dot{W}_{i,t+1}^h$.
 - (e) Recover $\dot{w}_{i,t+1}^h$ from (8) and $\dot{r}_{i,t+1}$ from (9).
 - (f) Iterate (b)–(e) with damping until $\max |\dot{p}_{i,t+1}^{h,(k+1)} - \dot{p}_{i,t+1}^{h,(k)}| < 10^{-8}$.
4. Update aggregate price $\dot{p}_{i,t+1}$ via (5); recover level $p_{i,t+1} = p_{it}\dot{p}_{i,t+1}$.
5. Update investment return $R_{i,t+1}$ via (11).
6. Update capital stock $k_{i,t+2} = (1 - \varsigma_{i,t+1}^{(n)})R_{i,t+1}k_{i,t+1}$ via (10).
7. Update weights $\omega_{ji,t+1}^{h,o}$, $\phi_{i,t+1}^h$, trade shares $S_{ij,t+1}^h$.

Backward pass. For $t = T - 1, T - 2, \dots, 0$:

1. Compute proposed $\varsigma_{it}^{\text{new}}$ via (13).
2. Compute proposed $(\dot{u}_{it}^h)^{\text{new}}$ via (12).

Damped update. For each (i, h, t) :

$$\dot{u}_{it}^{h,(n+1)} = \lambda(\dot{u}_{it}^h)^{\text{new}} + (1 - \lambda)\dot{u}_{it}^{h,(n)}, \quad \varsigma_{it}^{(n+1)} = \lambda\varsigma_{it}^{\text{new}} + (1 - \lambda)\varsigma_{it}^{(n)}, \quad (\text{A.25})$$

with damping parameter $\lambda \in [0.3, 0.5]$.

Convergence. Iterate until

$$\max_{i,h,t} \left| \dot{u}_{it}^{h,(n+1)} - \dot{u}_{it}^{h,(n)} \right| < 10^{-8} \quad \text{and} \quad \max_{i,t} \left| \varsigma_{it}^{(n+1)} - \varsigma_{it}^{(n)} \right| < 10^{-8}. \quad (\text{A.26})$$

Because levels w_{i0}^h, p_{i0} are pinned down in calibration, subsequent levels inherit the scale automatically via $w_{it}^h = w_{i0}^h \prod_{s=1}^t \dot{w}_{is}^h$; no per-period re-normalization is required.

E Supplementary Tables and Figures

Table A.1. Province List and Regional Classification

ID	Province	Region	Notes
1	Beijing	East	Municipality
2	Tianjin	East	Municipality
3	Hebei	Central	
4	Shanxi	Central	
5	Inner Mongolia	West	
6	Liaoning	Northeast	
7	Jilin	Northeast	
8	Heilongjiang	Northeast	
9	Shanghai	East	Municipality
10	Jiangsu	East	
11	Zhejiang	East	
12	Anhui	Central	
13	Fujian	East	
14	Jiangxi	Central	
15	Shandong	East	
16	Henan	Central	
17	Hubei	Central	
18	Hunan	Central	
19	Guangdong	East	
20	Guangxi	West	
21	Hainan	East	
22	Chongqing	West	Municipality
23	Sichuan	West	
24	Guizhou	West	
25	Yunnan	West	
26	Shaanxi	West	
27	Gansu	West	
28	Qinghai	West	
29	Ningxia	West	
30	Xinjiang	West	

Table A.2. Robustness of Long-Run Welfare Gains

Specification	Aggregate CEV	East	Central	West	Northeast	Network share	Migration share	Capital share
Baseline	9.40	6.42	10.73	12.31	7.08	23.5	32.3	21.9
$\theta = 3$	8.12	5.88	9.17	10.84	6.49	19.4	31.1	22.6
$\theta = 8$	10.86	7.03	12.54	14.21	7.91	26.8	33.7	20.4
$\rho = 3$	10.21	6.78	11.91	13.44	7.63	22.1	38.6	19.2
$\rho = 8$	8.57	6.11	9.62	11.08	6.74	24.0	26.9	23.5
$\delta = 0.06$	9.77	6.58	11.21	12.86	7.33	22.7	31.4	24.6
$\delta = 0.10$	9.08	6.31	10.29	11.84	6.91	23.8	32.9	19.7
No high-speed rail after 2008	5.21	3.84	5.98	6.77	4.11	21.6	29.4	20.1
Shift-share railway attribution	8.63	6.02	9.81	11.42	6.58	22.8	31.7	21.4
Equal exposure weights	9.02	6.37	10.16	11.88	6.96	23.1	31.9	22.2
No input-output linkages	7.19	5.34	8.03	9.11	5.74	—	34.1	23.8
No migration	6.36	5.21	7.03	7.89	5.46	26.9	—	27.4
Fixed capital	7.34	5.77	8.19	9.07	6.02	25.1	36.8	—

Notes: All entries are percent welfare gains unless otherwise noted. Regional gains are employment-weighted CEVs. Network, migration, and capital shares are the percentage contributions of each channel to the aggregate long-run gain. The table is rotated because the robustness exercise reports many model variants.